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A Comparative Study Between Different Fuzzy-Based Nonlinear Controllers for UAV Coaxial Octorotor

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Abstract This paper presents and compares three fuzzy-based nonlinear control strategies: fuzzy PD controller (FPDC), sliding-mode PID (SMC-PID), and fuzzy sliding-mode controller (FSMC) for the altitude and attitude regulation of a six-degree-of-freedom (6-DOF) coaxial octorotor UAV. The proposed FSMC integrates a type-1 Mamdani fuzzy inference system in the switching law of SMC to suppress chattering while maintaining robustness and stability. A full 6-DOF dynamic model is used to evaluate controller's performances under rectangular and circular trajectories, with and without external wind disturbances. Quantitative evaluation using ITAE, IAE, and control effort indices shows that FSMC achieves the lowest tracking error and energy consumption compared with FPDC and SMC-PID. The obtained results confirm that FSMC provides superior tracking accuracy, robustness, and energy efficiency, which makes it a promising approach for real-time UAV applications.

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Keywords Coaxial octorotor · Dynamic model · Control · Coaxial octorotor · Sliding mode · Fuzzy logic · PID

Abbreviations

UAV	Unmanned aerial vehicle
6-DOF	Six degrees of freedom
MIMO	Multi-input–multi-output
SMC-PID	Sliding-mode PID controller
FSMC	Fuzzy sliding-mode controller
SMC	Sliding-mode controller
FPDC	Fuzzy proportional–derivative controller
F	The rotor force
u	The moment vector
φ	The roll angle
ψ	The yaw angle
θ	The pitch angle
T	The transformation matrix
R	The rotation matrix
Sgn	Signum function
m	Drone mass
l	Arm length
b	Lift coefficient
d	Drag coefficient
$K_1,\ K_2,\ K_3$	Transitional drag coefficients
$K_4,\ K_5,\ K_6$	Friction coefficients
$I_x,\ I_y,\ I_z$	Axis inertia

1 Introduction

A multirotor UAV is an autonomous aerial system equipped with more than two rotors. Several configurations (such as birotor, trirotor, quadrotor, hexarotor, and octorotor) are

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used in practice. Multirotor UAVs offer several advantages, including vertical take-off and landing (VTOL), hovering ability, and strong resistance to external disturbances. As a result, they have been widely adopted in fields such as aerial photography, precision agriculture, search and rescue operations, and military applications [1].

Among them, octorotor UAVs stand out due to their superior payload capacity since they can carry heavier loads more efficiently than other multirotor types. They are categorized into two main configurations: star and coaxial. In the star configuration, eight DC motors are positioned at distinct points, whereas in the coaxial design, the motors are arranged in pairs along the same axis. Octorotor UAVs offer numerous benefits, such as VTOL capability and stable hovering, while also exhibiting greater resilience to environmental disturbances compared to other drone types. Consequently, their use has been expanding rapidly across multiple industries [2]. Octorotor UAVs play a crucial role in various fields, particularly in research, where they contribute to advancements in artificial intelligence, robust control systems, and the development of intelligent control frameworks. Despite the multidomain applications advantage, controlling a coaxial octorotor remains a challenging task because of its nonlinear and strongly coupled six-degree-of-freedom dynamics. The system exhibits multiple interacting subsystems (altitude, pitch, roll, yaw, and horizontal motion) that are sensitive to parameter variations and external disturbances such as wind gusts. Designing a controller that ensures accurate trajectory tracking while maintaining robustness and energy efficiency is a key research problem addressed in this work.

Octorotors, as nonlinear multi-input multi-output (MIMO) dynamic systems, exhibit significant uncertainties that complicate their control and making stability assurance a critical challenge. To overcome these complexities, researchers have extensively explored advanced control techniques, including both conventional nonlinear controllers and intelligent control approaches. Additionally, hybrid control strategies that integrate artificial intelligence with traditional methods have gained considerable attention within the research community for their potential to enhance system performance and adaptability.

The control strategies for coaxial octorotors closely resemble those of quadrotors. Consequently, the extensive literature on quadrotor path tracking and stability provides a strong foundation for advancing coaxial octorotor control research [3–5,9–13]. Studies dedicated to coaxial octorotor control, such as [6,8–10,13], consider dynamic models of coaxial octorotors using the Newton–Euler method, capturing both position and attitude dynamics. These models serve as the foundation for designing controllers capable of addressing the system inherent nonlinearities and high coupling effects. Several control strategies have been proposed for coaxial octorotors, ranging from fuzzy logic and sliding-mode control to backstepping and hybrid approaches incorporating neural networks. Among these, some novel methods show significant promise in mitigating the inherent control challenges of coaxial octorotors.

Hybrid approaches have proven highly effective in overcoming control challenges. In [3], a PID sliding-mode fuzzy logic controller was proposed, where the PID controller acted as a switching surface and was auto-tuned using gradient descent techniques, while fuzzy logic minimized chattering effects. Likewise, Reference [4] explored metaheuristic optimization (cuckoo optimization and particle swarm approaches) in an exponential sliding-mode controller integrated with an adaptive fuzzy type-2 neural network. Moreover [9], optimization methods such as genetic algorithms and particle swarm optimization were employed to fine-tune parametric gains in sliding mode controllers. Similarly, Reference [11] the combined used of fuzzy logic with backstepping controllers, while Reference [12] implemented a backstepping fast terminal sliding-mode controller coupled with an extended inverse multi-quadratic radial basis function neural network for disturbance observation. Further advancement is presented in [14] where an adaptive backstepping sliding-mode controller was developed with optimized gains using the Levenberg–Marquardt algorithm, ensuring stability through Lyapunov theory. Similarly, Reference [15] introduced a fault-tolerant fuzzy sliding-mode controller that employed a radial basis function neural network (RBFNN) to estimate control laws without relying on inverse dynamics.

Other hybrid approaches combine fuzzy sliding-mode controllers with adaptive interval type-2 fuzzy logic or integrate robust backstepping sliding-mode controllers with adaptive RBFNNs to effectively compensate for system nonlinearities and external disturbances [7]. Additionally, Reference [16] proposed a fault-tolerant control strategy based on adaptive global fast dynamic sliding-mode techniques, augmented with fuzzy logic for uncertainty estimation. On the other hand, Reference [17] proposes a novel multiple-input–multiple-output (MIMO) super-twisting sliding-mode controller (SMC) that uses a Multilayer Perceptron Neural Network (MLPNN) to estimate uncertainties and disturbances. The stability of the closed-loop system is ensured through Lyapunov stability theory, which also facilitates the formulation of real-time update laws for the MLPNN weights.

Control methodologies for quadrotors have also seen significant advancements. References [18–24] have extensively explored robust and hybrid control techniques, including sliding-mode control and fuzzy logic. In [18], a back propagation neural network is used to adapt the sliding-mode surface in response to external disturbances, ensuring stability through Lyapunov theory. Similarly, Reference [25] applied a fuzzy logic-based approach for efficient octorotor

Table 1 Comparison of recent hybrid control methods for multirotor UAVs (2021–2025)

Reference and year	Controller type	System	Performance focus	Main limitation
[16] 2021	Adaptive Fuzzy Terminal SMC	Octorotor	Fault Tolerance	No explicit chattering suppression
[17] 2024	Adaptive Super-Twisting SMC	Coaxial Octorotor	Robustness	High control energy
[30] 2024	Model Predictive Control	Coaxial Octorotor	Dynamic Coupling	High computational cost
Our work	**Fuzzy Sliding-Mode Control (FSMC)**	**Coaxial Octorotor**	**Robustness, low chattering, low energy**	**Novel fuzzy-based switching law**

control, while Reference [26] introduced a combined fuzzy PID controller for attitude steering.

Table 1 compares representative hybrid control strategies for multirotor UAVs reported between 2021 and 2024. As observed, most existing fuzzy or adaptive SMC schemes still suffer from residual chattering or high computational cost, particularly for coaxial octorotor configurations. The proposed FSMC achieves smoother control action with lower energy demand while maintaining robustness.

Moreover, advancements in fault-tolerant control strategies have been noteworthy. Studies such as [27,28] have proposed self-tuning, adaptive, and multiplexing fault-tolerant control methods for UAVs, emphasizing the importance of accounting for uncertainties and perturbations in real-time operations.

Recent studies have explored hybrid and intelligent control strategies to address these issues. Works such as [16,17,30] have proposed adaptive fuzzy, fault-tolerant, and model-predictive schemes for multirotor UAVs. However, most existing methods still suffer from persistent chattering or high computational cost, particularly when applied to complex coaxial octorotor configurations. These limitations motivate the development of a fuzzy-based sliding-mode framework that can maintain robustness while reducing chattering and control effort.

Although the recent studies in the dynamic event triggering fuzzy adaptive control focused on the multi-agent systems such as [36–38], highlighting the high relevance to the multirotor control [13]. The coaxial octocopter's subsystems pitch, roll, yaw, altitude, x, and y interact dynamically with other subsystems, forming a network of coupled subsystems like the multi-agent systems (MAS). In this case, using an event-triggered fuzzy adaptive technique makes the computation and actuator activity less, the same as how event-triggered communication makes transmissions in MAS less. Also, the fact that octocopters have actuator saturation and nonlinear aerodynamic coupling is similar to the output limits that MAS has to deal with. Consequently, the control architectures developed for multi-agent coordination provide a foundational inspiration for the hybrid fuzzy sliding-mode control (FSMC) structure proposed in this work. This approach specifically adopts principles of robustness, online adaptability, and energy-efficient actuation under prescribed output constraints.

In this paper, we consider the nonlinearities and strong coupling effects inherent in the dynamic models of a coaxial octorotor by implementing and analyzing fuzzy-based nonlinear controllers for a six-degree-of-freedom (6-DOF) system. The proposed control framework is validated through simulations considering multiple desired trajectories in simulation using a MATLAB/Simulink-based dynamic model of the coaxial octorotor. Furthermore, this paper introduces the design and simulation of robust and adaptive control strategies tailored specifically for coaxial octorotors. Building upon established techniques, we explore novel controller combinations to address the unique dynamic challenges of these systems, contributing to the expanding research on multirotor UAV control. Finally, a comprehensive performance comparison of the proposed controllers regarding their effectiveness and stability is presented.

The main contributions and innovations of this paper are summarized as follows:

(1) **Novel FSMC Structure:** We designed a dedicated type-1 Mamdani fuzzy logic controller (FLC) that directly replaces the discontinuous switching term of the sliding-mode controller (SMC). This single-input–single-output (SISO) fuzzy inference system takes the sliding surface as its input and outputs a continuous control signal, thereby eliminating chattering without introducing a boundary layer that could compromise robustness. This structure provides a new smooth-switching mechanism specifically tailored for the nonlinear 6-DOF coaxial octorotor dynamics.

(2) **Comprehensive Comparative Study:** The paper presents a thorough comparative analysis of three distinct control strategies: fuzzy PD controller (FPDC), sliding-mode PID (SMC-PID), and the proposed fuzzy sliding-mode controller (FSMC). The controllers are evaluated not only in terms of tracking performance (ITAE and IAE

indices) but also regarding control effort and robustness under external disturbances. This multi-criteria comparison provides a more practical assessment for UAV control applications.

(3) **Systematic Validation under Realistic Scenarios:** The proposed control schemes were validated in MATLAB/Simulink through multiple trajectory-tracking scenarios (rectangular and circular paths) and subjected to external wind disturbances. These experiments provide a holistic evaluation of the controllers' performance, demonstrating the FSMC's superior stability, robustness, and energy efficiency.

(4) **Quantitative Performance Evaluation:** A detailed quantitative evaluation using ITAE, IAE, and control effort indices has been included to highlight the FSMC's superior tracking accuracy, robustness against disturbances, and lower energy consumption compared to FPDC and SMC-PID. These results further confirm the practical efficiency of the proposed hybrid controller.

Unlike existing fuzzy SMC hybrid schemes mainly designed for quadrotors, this study introduces a fuzzy-based switching law specifically adapted for a nonlinear 6-DOF coaxial octorotor. The proposed type-1 Mamdani fuzzy inference system directly substitutes the discontinuous SMC term, thus reducing chattering while preserving robustness and low energy consumption.

The remainder of the paper is organized as follows: Sect. 2 introduces the dynamic modeling of the target 6-DOF coaxial octorotor. Section 3 describes the development of the proposed controllers and provides details on the design of the hybrid controllers as well as the tuning of the controller parameters. Section 4 discusses the simulation results of the proposed controllers. Section 5 provides a comparison study on the applied controllers and their performances. Section 6 draws the main conclusions and proposes future research paths.

2 Coaxial Octorotor Dynamic Modeling

The dynamic model of the considered octorotor is based on the models presented in [21,29]. The coaxial octorotor dynamics is the same as the quadrotor, where the coaxial octorotor is better than the star octorotor in stability and size. In addition, a classical octorotor needs the addition of four arms that are taller than the other ones to ensure the separation between the used rotors. Figure 1 shows an example of a coaxial octorotor, which is an Onyx Star.

2.1 Coaxial Octorotor Configuration

The coaxial octorotor is a multirotor UAV that has eight rotors. The coaxial octorotor can be considered a quadrotor, but in each corner, there are two coaxial rotors, and each one has an opposite direction of rotation. The dynamics of the octorotor are derived using two inertia frames [X_e, Y_e, Z_e] and a body frame F_b [X_b, Y_b, Z_b] to describe the rotation and the movement of the coaxial octorotor. Also, this UAV has six degrees of freedom because it has three movement axis: X, Y, and Z, and three rotations about the X, Y, and Z axis called, respectively, pitch, roll, and yaw [30]. Figure 2 describes the configuration and frames used to model this UAV. This configuration gives the coaxial octorotor the ability of vertical landing and hovering.

Fig. 1 Onyx Star coaxial octorotor

2.2 Coaxial Octorotor Dynamics

The dynamic modeling of the octorotor leads to a MIMO nonlinear system with high coupling which is derived using Newton–Euler method following the below hypothesis to obtain the dynamic model [21,29,31,32]:

- The octorotor is supposed as a rigid body and symmetrical.
- The center of mass of the octorotor and the origin of the body frame are supposed to coincide.
- The rotors are supposed rigid.
- The lift and drag forces are proportional to the square of the angular velocity.

The octorotor vehicle has six degrees of freedom (DOF) because it can do a vertical movement in *x, y* axis and an horizontal motion in *z* axis. In addition, it can roll and pitch about *x, y* and yaw in *z* axis. These movements are done in the body frame F_b which is aligned to the center of mass of the octorotor. For that, in order to describe these movements in the inertia F frame, we need to transform these movements from the F_e to F_b. This transformation is done using the following transformation matrix T [10,30,32]:

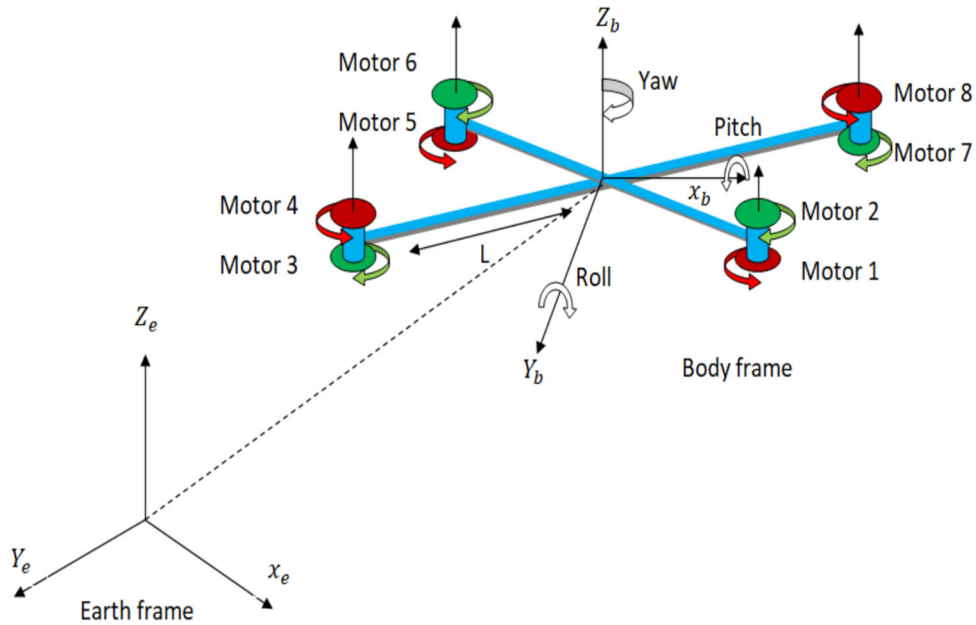


Fig. 2 Coaxial octorotor configuration

$$T = \begin{bmatrix} R & \xi \\ 0 & 1 \end{bmatrix}, \tag{1}$$

$$R = \begin{bmatrix} C_\theta C_\psi & S_\varphi S_\theta S_\psi - C_\varphi C_\psi & C_\varphi S_\theta C_\psi + S_\varphi S_\psi \\ C_\theta S_\psi & S_\varphi S_\theta S_\psi + C_\varphi C_\psi & C_\varphi C_\theta S_\psi - S_\varphi S_\psi \\ -S_\theta & S_\varphi C_\theta & C_\varphi C_\theta \end{bmatrix}, \tag{2}$$

where R is the rotation matrix which is derived using Euler angles flight notation [30–32] and $\xi = \begin{bmatrix} x & y & z \end{bmatrix}$ is the position vector.

The dynamic model of the octorotor is developed using the Newton–Euler formula.

$$\begin{cases} \dot{\xi} = v, \\ m\ddot{\xi} = F_g + F_L + F_v, \\ J\dot{\Omega} = -M_{gb} + \dot{M}_L - M_{gr} - M_f. \end{cases} \tag{3}$$

To obtain the equation that describes the position in the space, we have to consider that F_g is the gravity vector, F_L is the lift force matrix, and F_v is the drag force matrix leading to the following representation:

$$m \times \begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -gm \end{bmatrix} + R \times \begin{bmatrix} 0 \\ 0 \\ \sum_1^8 F_i \end{bmatrix} + \begin{bmatrix} K_1 & 0 & 0 \\ 0 & K_2 & 0 \\ 0 & 0 & K_3 \end{bmatrix} \times v, \tag{4}$$

where v is the velocity vector; $F_i = b\omega_i$, where b is the lift coefficient, and ω_i is the angular velocity for each rotor.

The equation describing the rotations is the following:

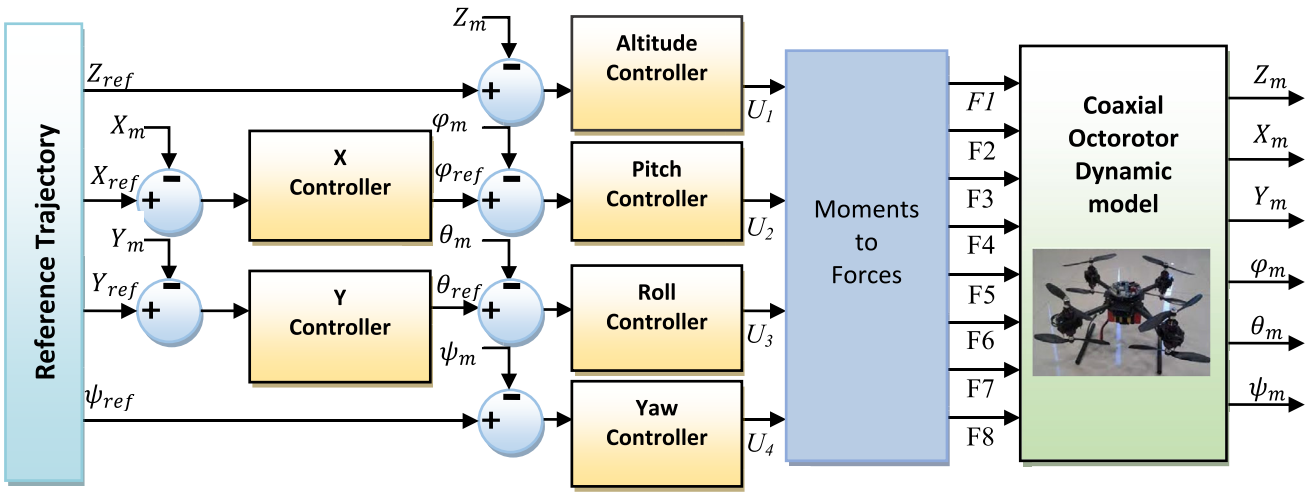
$$\begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix} \times \begin{bmatrix} \ddot{\varphi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} = -\Omega \Lambda J \Omega + \begin{bmatrix} U_2 \\ U_3 \\ U_4 \end{bmatrix} - \sum_{i=1}^{8} \Omega \Lambda J_r \begin{bmatrix} 0 & 0 & (-1)^{i+1}\omega_i \end{bmatrix} + \begin{bmatrix} K_4 & 0 & 0 \\ 0 & K_5 & 0 \\ 0 & 0 & K_6 \end{bmatrix} \times \begin{bmatrix} \varphi^2 \\ \theta^2 \\ \psi^2 \end{bmatrix}, \tag{5}$$

$$\begin{cases} \ddot{x} = u_x - \frac{K_1}{m}\dot{x}, \\ \ddot{y} = u_y - \frac{K_2}{m}\dot{y}, \\ \ddot{z} = \frac{\cos\varphi\cos\theta}{m}u_1 - \frac{K_3}{m}\dot{z} - g, \\ \ddot{\varphi} = \frac{(I_y - I_z)}{I_x}\dot{\theta}\dot{\psi} - \frac{K_4}{I_x}\dot{\varphi}^2 - \frac{J_r\bar{\Omega}}{I_x}\dot{\theta} + \frac{1}{I_x}u_2, \\ \ddot{\theta} = \frac{(I_z - I_x)}{I_y}\dot{\varphi}\dot{\psi} - \frac{K_5}{I_y}\dot{\theta}^2 - \frac{J_r\bar{\Omega}}{I_y}\dot{\varphi} + \frac{1}{I_y}u_3, \\ \ddot{\psi} = \frac{(I_x - I_y)}{I_z}\dot{\theta}\dot{\varphi} - \frac{K_6}{I_z}\dot{\psi}^2 + \frac{1}{I_z}u_4, \end{cases} \tag{6}$$

where:

$$\begin{aligned} u_x &= \frac{u_1}{m}(\cos\varphi\sin\theta\cos\psi + \sin\varphi\sin\psi), \\ u_y &= \frac{u_1}{m}(\cos\varphi\sin\theta\cos\psi - \sin\varphi\sin\psi), \\ \bar{\Omega} &= [\omega_1 - \omega_2 + \omega_3 - \omega_4 + \omega_5 - \omega_6 + \omega_7 - \omega_8]. \end{aligned} \tag{7}$$

And u_1, u_2, u_3, u_4 represent the total thrust, pitch, roll, and yaw moments, respectively, that are defined as follows:



Strategy/controller	X	Y	Altitude	Pitch	Roll	Yaw
SMC-PID	PID	PID	SMC	SMC	SMC	SMC
FPDC	PD	PD	FLC	FLC	FLC	LC
FSMC	PI	PI	FSMC	FSMC	FSMC	FSMC

Fig. 3 The overall control methodology

$$
\begin{aligned}
u_1 &= (F_1 + F_2 + F_3 + F_4 + F_5 + F_6 + F_7 + F_8),\\
u_2 &= c\,(-F_1 - F_2 - F_3 - F_4 + F_5 + F_6 + F_7 + F_8)\,,\\
u_3 &= c\,(-F_1 - F_2 + F_3 + F_4 + F_5 + F_6 - F_7 - F_8)\,,\\
u_4 &= d\,(-F_1 + F_2 + F_3 - F_4 - F_5 + F_6 + F_7 - F_8)\,,
\end{aligned} \tag{8}
$$

where $c = l\frac{\sqrt{2}}{2}$ and d is the drag coefficient.
These moments are assembled in one vector U, which is considered as the input vector. The output vector Y is defined considering that all states are measured.

$$
\begin{aligned}
U &= \left[u_1\ u_2\ u_3\ u_4\right]^{\mathrm{T}},\\
Y &= \left[X\ Y\ Z\ \varphi\ \theta\ \psi\right]^{\mathrm{T}}.
\end{aligned} \tag{9}
$$

3 The Proposed Controllers

The coaxial octorotor is a high-coupled MIMO nonlinear system with uncertainties and external disturbances during motion. To ensure an effective guidance system, a stable and robust controller is required. In this section, the proposed control structures using different controller types are illustrated and discussed. We have used different techniques (such as SMC and fuzzy Logic) to guide and stabilize the coaxial octorotor.

3.1 Overall Control Structure

The overall control architecture of the coaxial octorotor is organized into four main subsystems: altitude (z), position (X and Y), and attitude [pitch (θ), roll (φ), and yaw (ψ)]. The altitude and yaw motions are each regulated by dedicated controllers, whereas the pitch and roll subsystems form the inner attitude-control loop, with the $X-Y$ translational motion forming the outer position-control loop. Each subsystem operates in coordination within the hierarchical structure shown in Fig. 3. This figure presents the generalized control framework for all proposed fuzzy-based nonlinear controllers (FPDC, SMC-PID, and FSMC), illustrating the flow of reference inputs, error generation, and control actions applied to the octorotor dynamics. A table legend is associated to illustrate the employed strategy/controller for different variables (X, Y, altitude, pitch, roll, and yaw).

The detailed design of each control strategy will be described in the following sections as depicted in Figs. 4, 8, and 11.

The dynamic model of the octorotor, derived in Sect. 2, serves as the foundation for the model-based controllers. The relevant dynamics for the controlled subsystems are as follows:

– **Altitude dynamics:**

$$
\ddot{z} = \frac{\cos\varphi\cos\theta}{m}u_1 - \frac{K_3}{m}\dot{z} - g. \tag{10}
$$

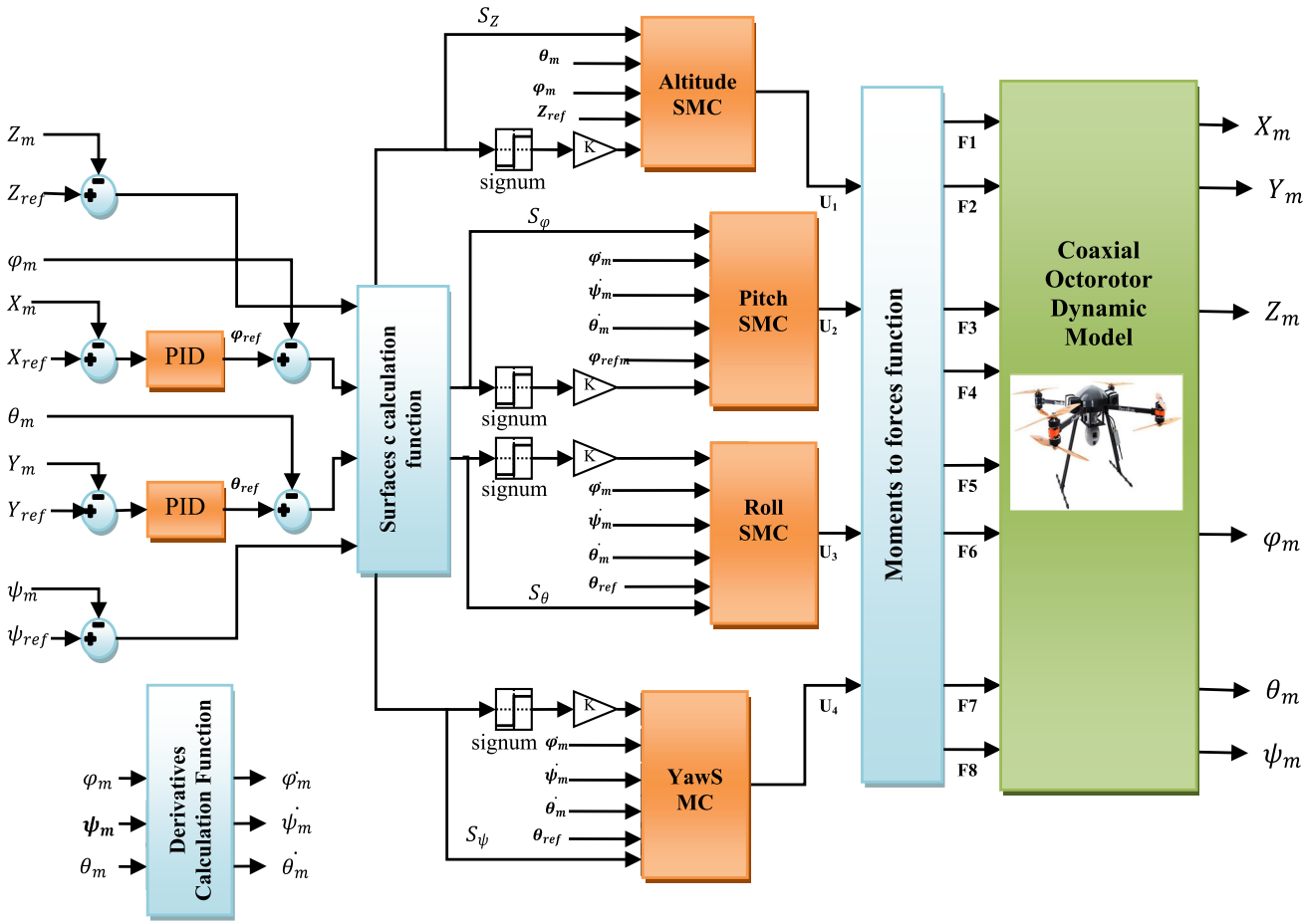


Fig. 4 SMC-PID control structure for the path-tracking coaxial octorotor

– **Roll (φ) dynamics:**

$$\ddot{\varphi} = \frac{(I_y - I_z)}{I_x}\dot{\theta}\dot{\psi} - \frac{K_4}{I_x}\dot{\varphi}^2 - \frac{J_r\bar{\Omega}}{I_y}\dot{\varphi} + \frac{1}{I_x}u_2 \tag{11}$$

– **Pitch (θ) dynamics:**

$$\ddot{\theta} = \frac{(I_z - I_x)}{I_y}\dot{\varphi}\dot{\psi} - \frac{K_5}{I_y}\dot{\theta}^2 - \frac{J_r\bar{\Omega}}{I_x}\dot{\theta} + \frac{1}{I_y}u_3. \tag{12}$$

– **Yaw (ψ) dynamics:**

$$\ddot{\psi} = \frac{(I_x - I_y)}{I_z}\dot{\theta}\dot{\varphi} - \frac{K_6}{I_z}\dot{\psi}^2 + \frac{1}{I_z}u_4. \tag{13}$$

For all controllers, the desired pitch (θ_d) and roll (φ_d) angles are generated from the X and Y position errors using conventional PID controllers as follows:

– **Desired pitch:**

$$\theta_d = K_p \cdot e_y + K_i \cdot \int e_y + K_d \cdot \frac{de_y}{dt} \tag{14}$$

– **Desired roll:**

$$\varphi_d = K_p \cdot e_x + K_i \cdot \int e_x + K_d \cdot \frac{de_x}{dt}, \tag{15}$$

where $e_x = x_d - x$, $e_y = y_d - y$

For the sliding-mode control (SMC)-based strategies (SMC-PID and FSMC), the sliding surfaces for the subsystems are defined as follows:

$$S_Z = \dot{e}_Z + \lambda_Z e_Z S_\phi = \dot{e}_\phi + \lambda_\phi e_\phi, \tag{16}$$

$$S_\theta = \dot{e}_\theta + \lambda_\theta e_\theta S_\psi = \dot{e}_\psi + \lambda_\psi e_\psi, \tag{17}$$

where $e_z = z_d - z_e$, $e_\phi = \phi_d - \phi$, $e_\theta = \theta_d - \theta$, and $e_\psi = \psi_d - \psi_e$.

3.2 Sliding-Mode PID Controller (SMC-PID)

This controller applies a first-order sliding-mode controller to the altitude, roll, pitch, and yaw subsystems. The key charac-

teristic of this approach is the use of the discontinuous signum function, sgn(S), which provides robustness but introduces high-frequency oscillations known as chattering.

The control laws are derived from the dynamics in Sect. 3.1 and are given by:

- **Altitude control (U_1):**

$$U_1 = \frac{1}{\cos\varphi \cdot \cos\theta} \left[K_3 + m_g + \ddot{z}_d \cdot m + \dot{e} \cdot c \cdot m + \text{sgn}(s) \cdot K\right] \quad (18)$$

- **Roll control (U_2):**

$$U_2 = I_x \left[\dot{e}_\varphi \cdot c + \ddot{\varphi}_d - \left(\dot{\theta} \cdot \dot{\psi} \cdot A_1 - \dot{\varphi}^2 \cdot A_2 - A_3 \cdot \dot{\varphi}\right) + \text{sgn}(s) \cdot K\right]. \quad (19)$$

- **Pitch control (U_3):**

$$U_3 = I_y \left[\dot{e}_\theta \cdot c + \ddot{\theta}_d - \left(\dot{\varphi} \cdot \dot{\psi} \cdot A_5 - \dot{\theta}^2 \cdot A_6 - A_7 \cdot \dot{\theta}\right) + \text{sgn}(s) \cdot K\right]. \quad (20)$$

- **Yaw control (U_4):**

$$U_4 = I_z \left[\dot{e}_\psi \cdot c + \ddot{\psi}_d - \left(\dot{\varphi} \cdot \dot{\theta} \cdot A_9 - \dot{\psi}^2 \cdot A_8\right) + \text{sgn}(s) \cdot K\right]. \quad (21)$$

The control structure for the SMC-PID is presented in Fig. 4.

3.3 Fuzzy Proportional–Derivative Controller (FPDC)

The second control strategy is a fuzzy logic controller (FLC) for the altitude, roll, pitch, and yaw subsystems. This is a model-free approach that relies on expert knowledge.

Each FLC has two inputs: the error (e) and the derivative of the error (de). The output is the control action.

- The input e uses five trapezoidal membership functions: Negative Big (NB), Negative (N), Near Zero (NZ), Positive (P), and Positive Big (PB), as shown in Fig. 5.
- The input de uses three Gaussian membership functions: Negative (N), Zero (Z), and Positive (P), as shown in Fig. 6.
- The output uses five membership functions: Go Much Down (GDM), Go Down (GD), Stay (S), Go Up (GU), and Go Up Much (GUM), as shown in Fig. 7.
- The crisp output is inferred using the centroid method. The fuzzy rule base is summarized in Table 2.

The choice of five trapezoidal membership functions for the error (e) was made to provide a wide operating range

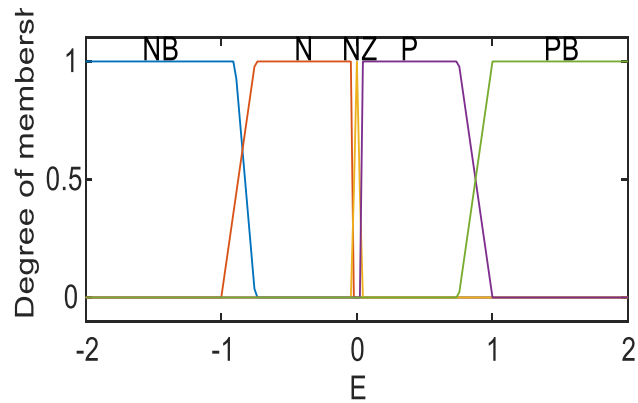


Fig. 5 Input variable membership functions (e)

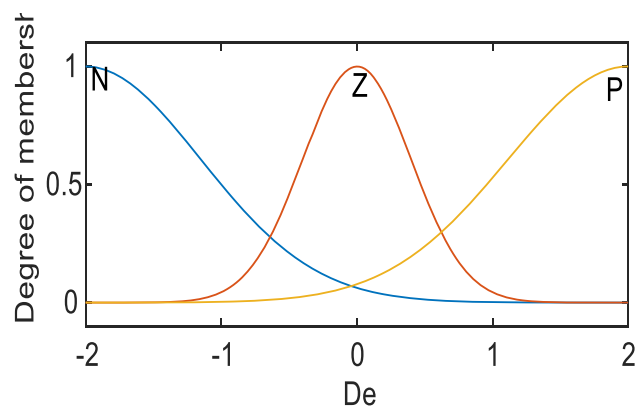


Fig. 6 Membership functions of (de)

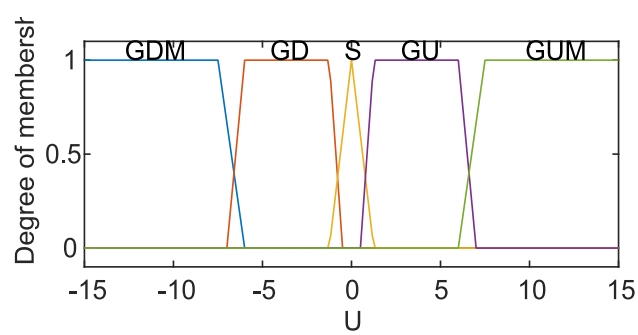


Fig. 7 Output variable membership functions

Table 2 Fuzzy rules

de	e	NB	N	NZ	P	PB
N		GDM	GD	GD	S	GU
Z		GUM	GD	S	GU	GUM
P		GD	S	GU	GUM	GUM

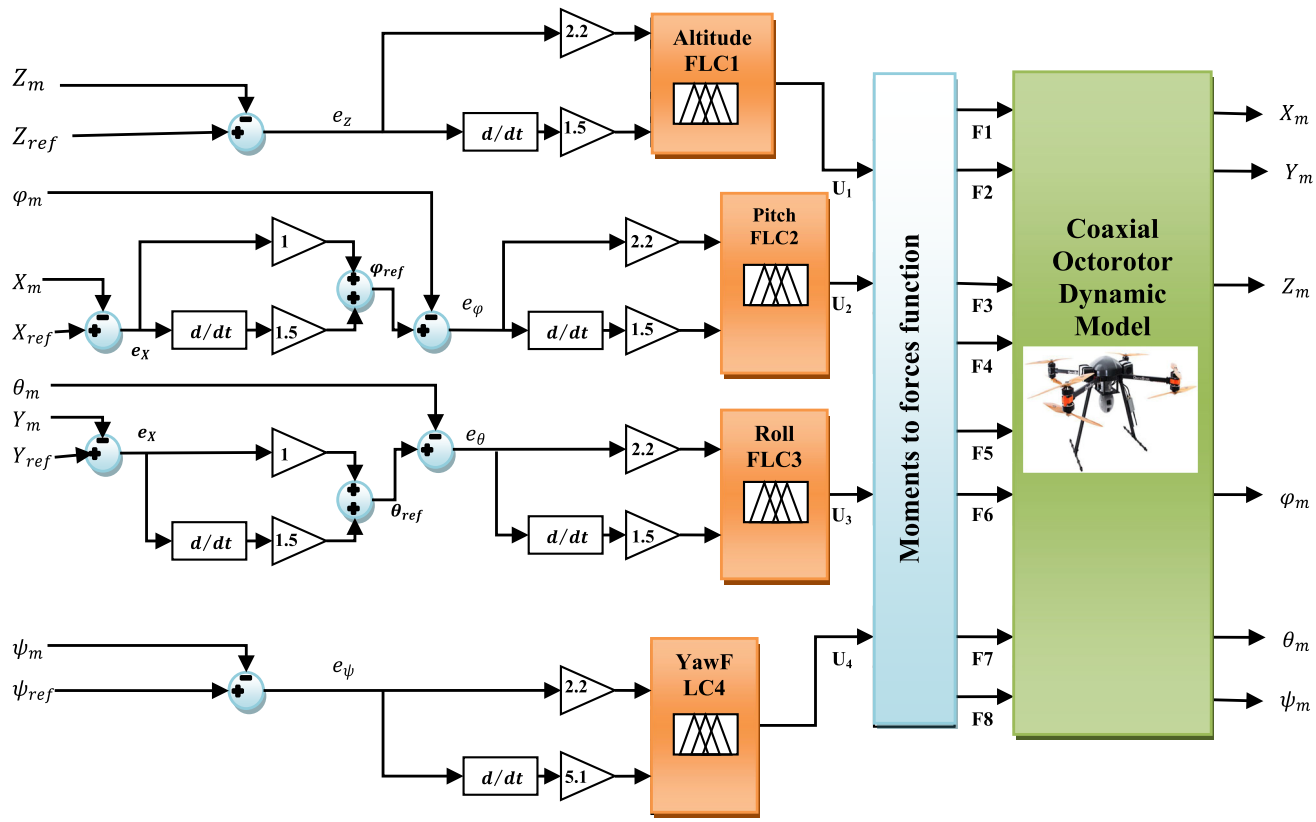


Fig. 8 FPD control structure for the coaxial octorotor

Table 3 Gains of the employed PD controllers

Controller	Gain	
	K_p	K_d
Altitude	2.2	1.5
Pitch	2.2	1.5
Roll	2.2	1.5
Yaw	2.2	5.1
X and Y	1	1.5

and a clear, intuitive interpretation of the system's state. Trapezoidal functions offer a constant, high sensitivity over a significant portion of the input domain, which is desirable for steady error correction. For the derivative of the error (de), three Gaussian membership functions were selected. The smooth transitions of Gaussian functions around zero are crucial for stabilizing the controller's output and preventing abrupt changes when the error derivative is near zero, thus contributing to a smoother system response.

The rule base (Table 2) was constructed based on a heuristic understanding of PD control. The fundamental principle is to combine the current error and its trend to determine the appropriate control action. For instance, the rule (***IF e is NB AND de is N THEN output is GDM)*** represents a situation where the system is far below the set point and moving further away, necessitating a large corrective action in the positive direction. This intuitive design allows for effective control without requiring an exact analytical model of the complex octorotor dynamics.

For the X and Y position subsystems, a proportional–derivative (PD) controller is employed to generate the desired pitch and roll values, with gains detailed in Table 3. The proposed control structure using FPDC is presented in Fig. 8.

3.4 Fuzzy Sliding-Mode Controller (FSMC)

The FSMC is a hybrid controller designed to leverage the robustness of SMC while mitigating its chattering effect. The key innovation is the replacement of the discontinuous sgn(S) term in the SMC-PID with a continuous output from a fuzzy inference system (FIS).

A type-1 Mamdani FIS with a single input (the sliding surface S) and a single output (the switching control signal U_S) is implemented. The input S uses five trapezoidal membership functions: BN (Big Negative), N (Negative), Z (Zero), P (Positive), and BP (Big Positive), as shown in Fig. 9. The output U_S uses five membership functions: GDM, GD, S, GU, and GUM, as shown in Fig. 10. The rule base is summarized in Table 4.

The final control inputs for each subsystem are given by:

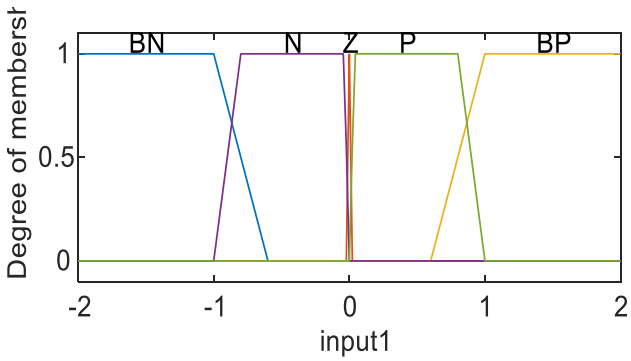


Fig. 9 The input membership functions

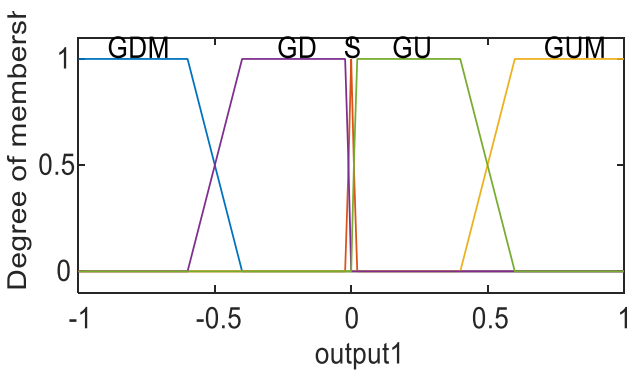


Fig. 10 The output membership function

Table 4 Fuzzy logic rule base

S_x	BN	N	Z	P	BP
Output	GDM	GD	S	GU	GUM

- **Altitude control (U_1)** :

$$U_1 = \frac{1}{\cos\varphi \cdot \cos\theta}\left[K_3 + m_g + \ddot{z}_d \cdot m + \dot{e} \cdot c \cdot m + U_{SZ}\right], \tag{22}$$

where $U_{SZ} = K_f(S_Z)^* K$.

- **Roll control (U_2)** :

$$U_2 = I_x\left[\dot{e}_\varphi \cdot c + \ddot{\varphi}_d - \left(\dot{\theta} \cdot \dot{\psi} \cdot A_1 - \dot{\varphi}^2 \cdot A_2\right) + U_{S\varphi}\right], \tag{23}$$

where $U_{S\varphi} = K_f(S_\varphi) * K$.

- **Pitch control (U_3)** :

$$U_3 = I_y\left[\dot{e}_\theta \cdot c + \ddot{\theta}_d - \left(\dot{\varphi} \cdot \dot{\psi} \cdot A_5 - \dot{\theta}^2 \cdot A_6\right) + U_{S\theta}\right], \tag{24}$$

where $U_{S\theta} = K_f(S_\theta)^* K$.

- **Yaw control (U_4)** :

$$U_4 = I_z\left[\dot{e}_\psi \cdot c + \ddot{\psi}_d - \left(\dot{\varphi} \cdot \dot{\theta} \cdot A_9 - \dot{\psi}^2 \cdot A_{10}\right) + U_{S\psi}\right], \tag{25}$$

where $U_{S\psi} = K_f(S_\psi) * K$

Here, $K_f(S)$ represents the output of the fuzzy inference system.

In the FSMC, the fuzzy system is designed with a single input (the sliding surface S) and a single output (the switching control signal (U_S). The five trapezoidal input membership functions (BN, N, Z, P, BP) were chosen to partition the state of the system relative to the sliding manifold. This provides a straightforward mapping: a large positive S requires a large positive control action to drive the system back to the manifold and vice versa.

The output membership functions (GDM, GD, S, GU, GUM) are designed to provide a smooth, continuous approximation of the ideal, discontinuous SMC law. The rule base is constructed to directly emulate the desired behavior of the SMC while eliminating chattering. For example, the rule ***IF S is BP THEN U is GUM*** ensures a strong control action when the system is far from the sliding surface, similar to the SMC's high-gain principle. However, as S approaches zero (rules for Z and N/P), the output smoothly reduces to zero, effectively creating an adaptive boundary layer that eliminates the high-frequency switching of the traditional ***sgn(S)*** function. This design choice is the core innovation that grants the FSMC its robustness without the detrimental chattering effect.

The proposed control structure for the FSMC is presented in Fig. 11.

– **Discussion on the Innovations of the Proposed FSMC**

The proposed fuzzy sliding-mode controller (FSMC) introduces several structural and theoretical innovations compared with previously developed hybrid control schemes such as classical fuzzy SMC and adaptive fuzzy–neural controllers.

Unlike many existing methods in which fuzzy logic merely adjusts switching gains or tunes SMC parameters, the proposed FSMC replaces the discontinuous switching term $[K_s \cdot \mathrm{sign}(S)]$ entirely with a continuous fuzzy inference signal $K_f(S)$.

This modification removes the need for an artificial boundary layer that can weaken robustness and ensures a smooth control signal that effectively eliminates chattering.

Structurally, the FSMC adopts a single-input single-output (SISO) fuzzy logic framework using only the sliding surface S as input.

This configuration simplifies computation and facilitates real-time implementation for the 6-DOF coaxial octorotor,

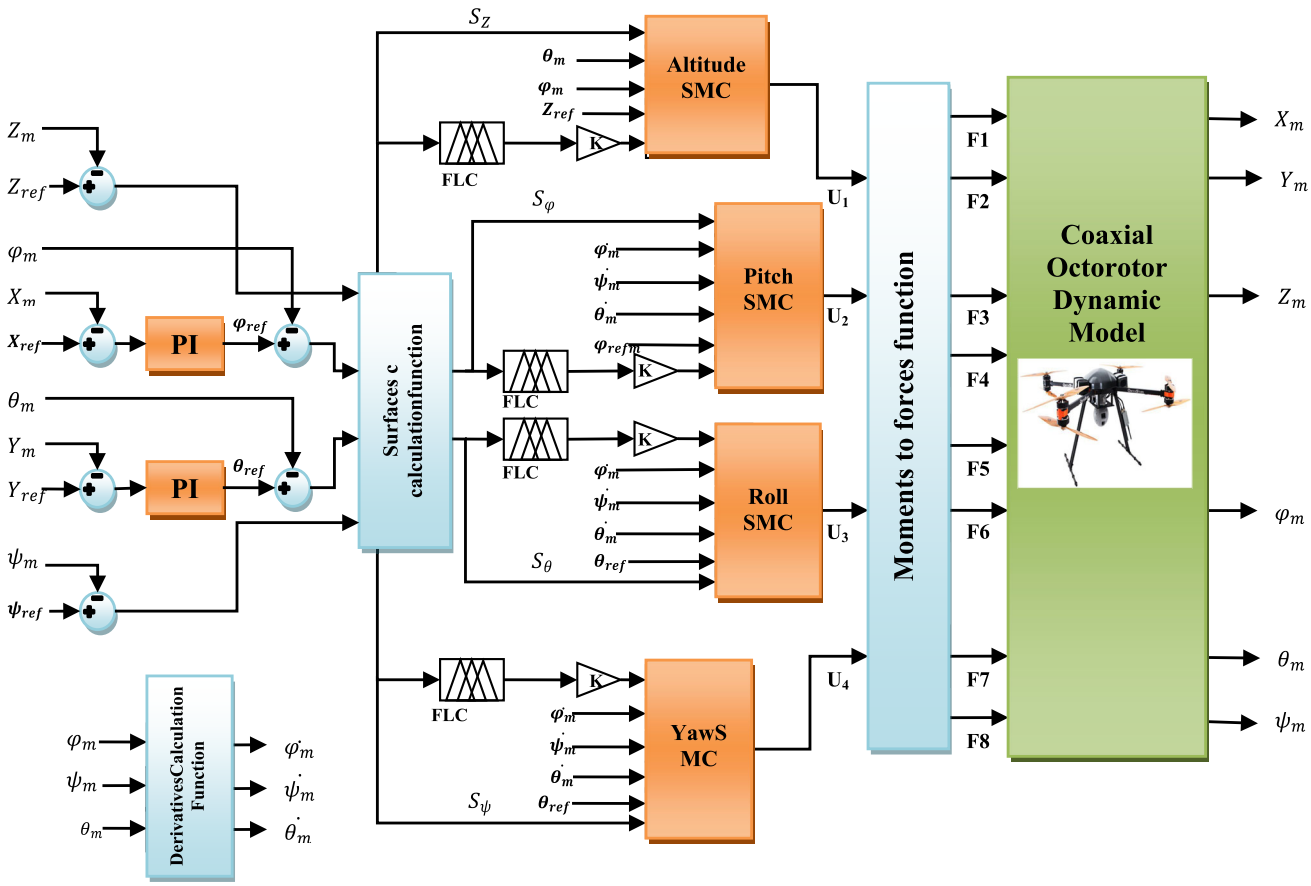


Fig. 11 The FSMC control structure for the coaxial octorotor

whose dynamics exhibit strong coupling between translational and rotational motions.

The fuzzy rule base and membership functions are designed so that large $|S|$ values yield strong corrective actions for rapid convergence, while small $|S|$ values near equilibrium produce softer control efforts, minimizing overshoot and improving energy efficiency. These choices distinguish the FSMC from other fuzzy SMC methods that rely primarily on gain adaptation or boundary-layer smoothing.

In addition, the proposed design maintains the theoretical stability properties of the conventional SMC. The fuzzy control signal preserves the reaching and sliding conditions required for convergence, thereby combining robustness and adaptability within a smooth nonlinear framework specifically optimized for coaxial octorotor dynamics.

– **Stability Analysis of the FSMC Closed-Loop System**

The stability of the proposed FSMC is demonstrated using Lyapunov theory such as:

Let the sliding surface be defined as:

$$S = \dot{e} + \lambda e, \tag{26}$$

where e denotes the tracking error and $\lambda > 0$ is a positive constant determining the desired convergence rate.

The Lyapunov candidate function is selected as:

$$V(s) = \frac{1}{s} s^2. \tag{27}$$

Taking its time derivative yields:

$$\tilde{V} = S \cdot S. \tag{28}$$

The FSMC control law can be expressed as:

$$u = u_{\text{eq}} + K_{\text{f}}(s), \tag{29}$$

where u_{eq} is the equivalent control that compensates nominal dynamics, and $K_{\text{f}}(S)$ is the fuzzy control output replacing the discontinuous term in classical SMC.

Substituting this control into the closed-loop error dynamics gives

$$\dot{s} = f(x, t) - u_{\text{eq}} - K_{\text{f}}(s). \tag{30}$$

Table 5 Parameters of the dynamic model

Parameter	Simulated Value
b	2.9842×10^{-5} N m/rad/s
ld	2.2320×10^{-7} N m/rad/s
K_1, K_2, K_3	$(5.5670,\ 5.5670,\ 6.3540) \times 10^{-2}$ N m/rad/s
K_4, K_5, K_6	$(5.5670,\ 5.5670,\ 6.3540) \times 10^{-2}$ N m/rad/s
J_r	2.8385×10^{-5} N m/rad/s^2
I_x, I_y, I_z	$(4.2,\ 4.2,\ 7.5) \times 10^{-2}$ N m/rad/s^2
m	1.6 kg
$4l$	0.23 m

If the fuzzy output satisfies the condition

$$s \cdot K_f(s) > 0. \tag{31}$$

Then

$$\tilde{V} = -s \cdot K_f(s) < 0. \tag{32}$$

Since $V(s)$ is positive definite and $\tilde{V}$ is negative definite, it follows that all system trajectories asymptotically converge to the sliding surface $S = 0$.

Therefore, the FSMC guarantees global asymptotic stability while producing a smooth control signal free from chattering. This theoretical result is consistent with the simulation outcomes presented in Sect. 4, confirming the robustness and effectiveness of the proposed control scheme.

4 Simulation Results and Discussion

In this section, we will present some simulation results for the evaluation of the proposed controllers using a MATLAB/Simulink simulation of a 6-DOF coaxial octorotor described by the dynamic model in Sect. 2. Table 4 shows the parameters of the dynamic drone model. The proposed controllers are evaluated regarding path-tracking performance, i.e., the ability of each controller to steer the octorotor drone toward the desired trajectory and to keep it near its reference coordinates (Table 5).

4.1 Path-Tracking Scenarios

In our simulation, two types of paths are used as reference trajectories.

a. Rectangular path

This reference path starts from the origin point to reach a maximum height of 2 m. After that, the coaxial octorotor should move a maximum of 2 m along the x-axis, and then it will reach a maximum of 2 m along the y-axis. While the yaw angle of the coaxial octorotor starts at 0 rad, it reaches 0.25 rad during the time range [20 s, 30 s] and 0.35 rad during the time range [30 s, 80 s]. On the other hand, the rotation finishes at 1 rad. These movements are described using step functions for the three coordinates (x, z, and ψ):

$$X_d = \begin{cases} 1; & t < 20, \\ 2; & 20 < t > 60, \\ 0.5; & t > 60, \end{cases} \tag{33}$$

$$Y_d = \begin{cases} 1; & t < 30, \\ 2; & 30 < t > 80, \\ 0.5; & t > 80, \end{cases} \tag{34}$$

$$Z_d = \begin{cases} 2; & t < 80, \\ 1; & t > 80, \end{cases} \tag{35}$$

$$\psi_d = \begin{cases} 0; & t < 20, \\ 0.25; & 20 < t > 30, \\ 0.35; & 30 < t > 80, \\ 1; & t > 80, \end{cases} \tag{36}$$

where X_d, Y_d, Z_d and ψ_d are the references for the rectangular trajectory.

b. Circular path

This reference path will force the coaxial octorotor start from the origin point, reaching a height of 3 m and a maximum of 2 m along the X-axis. At this point, the coaxial octorotor will start rotating, forming a circular path with a diameter of 2 m. The coaxial octorotor will then return to its starting point of rotation and descend straight to the origin. These movements are performed by specifying the following references

$$X_d = 4 \cdot \cos\left((t \cdot 0.24 - 1) \cdot \frac{\pi}{10}\right), \tag{37}$$

$$Y_d = 4 \cdot \sin\left((t \cdot 0.24 - 1) \cdot \frac{\pi}{10}\right), \tag{38}$$

$$Z_d = \begin{cases} 1; & t < 10, \\ 3; & t > 10, \\ 0; & t > 110, \end{cases} \tag{39}$$

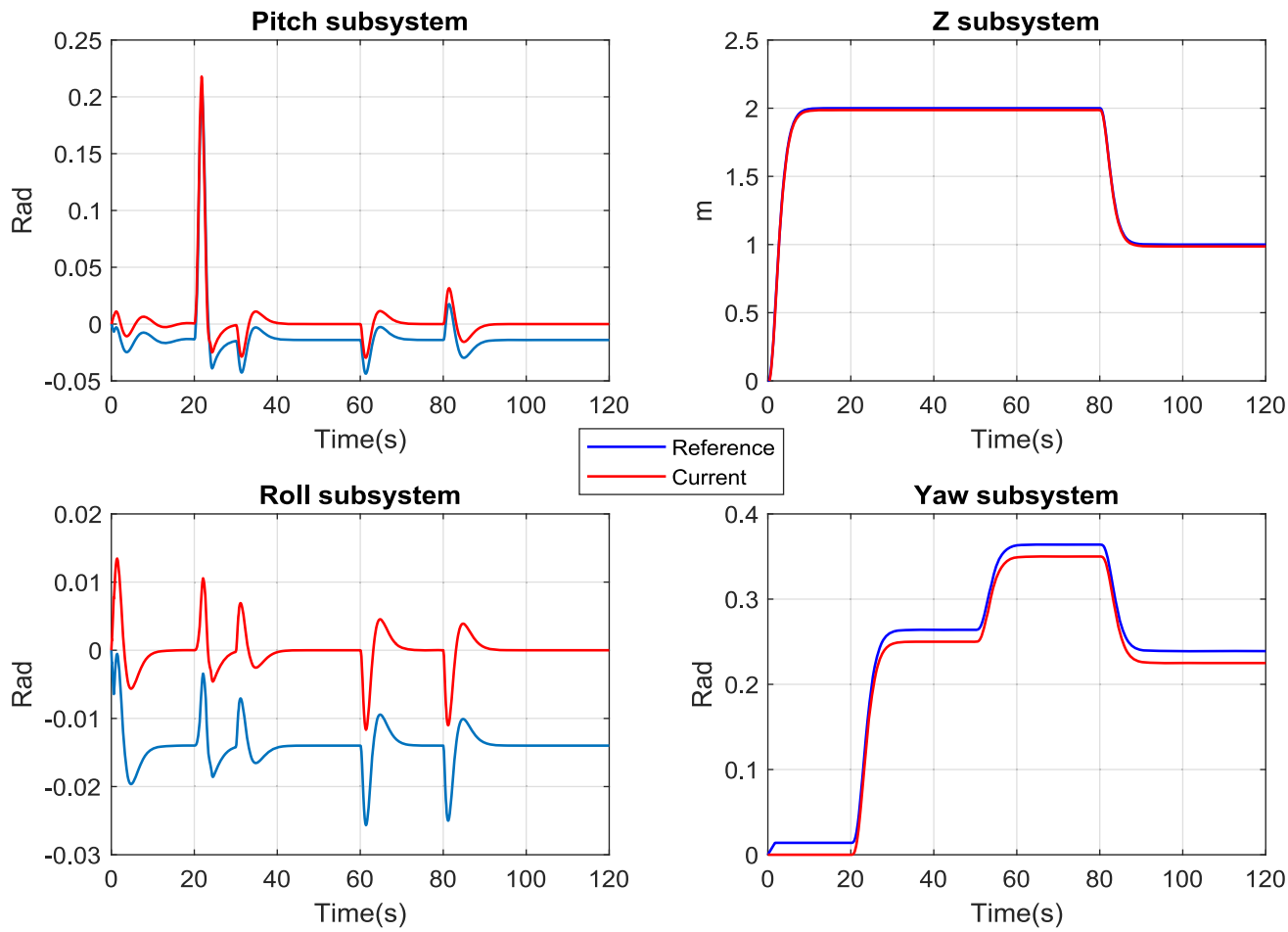


Fig. 12 SMC-PID subsystems' responses

where X_v, Y_v, and Z_d are the references for the circular trajectory.

Additionally, to the test the robustness of the controllers, an external wind disturbance in the pitch angle was applied on the coaxial octorotor model for each controller defined as follows:

$$d = \begin{cases} 0; & t < 40, \\ 0.2; & 40 < t < 42, \\ 1; & t > 40, \end{cases} \tag{40}$$

where d is the external disturbance.

4.2 Subsystems Responses and Path-Tracking Results Analysis

Figures 12, 13, 14, 15, 16, 17, and 18 presents the simulation results based on the application of the proposed controllers. The obtained results highlight some important characteristics of performance analysis, chattering reduction, path tracking, and robustness to external disturbances.

a. Subsystems responses:

Figures 12, 13, 14, 15, and 16 show the response and control signals of the pitch, roll, yaw, and altitude subsystems using SMC-PID, fuzzy PD, and FSMC controllers on a rectangular path-tracking scenario.

– **SMC-PID:**

Figure 12 shows that the SMC-PID controller retains stability and error tracking; nevertheless, the control signals generated by the SMC-PID controller exhibit the chattering phenomena in Fig. 15, which is a common problem in sliding-mode controllers. This problem is generated by a discontinuous switching signal, which can result in wasteful power usage and mechanical stress, generating vibration in the actuators and reducing their life lifetime.

– **Fuzzy PID:**

The fuzzy PID controller likewise maintains stability and has a lower steady-state error than the SMC-PID in Fig. 13, with smooth control signals in Fig. 16 due to the trapezoidal

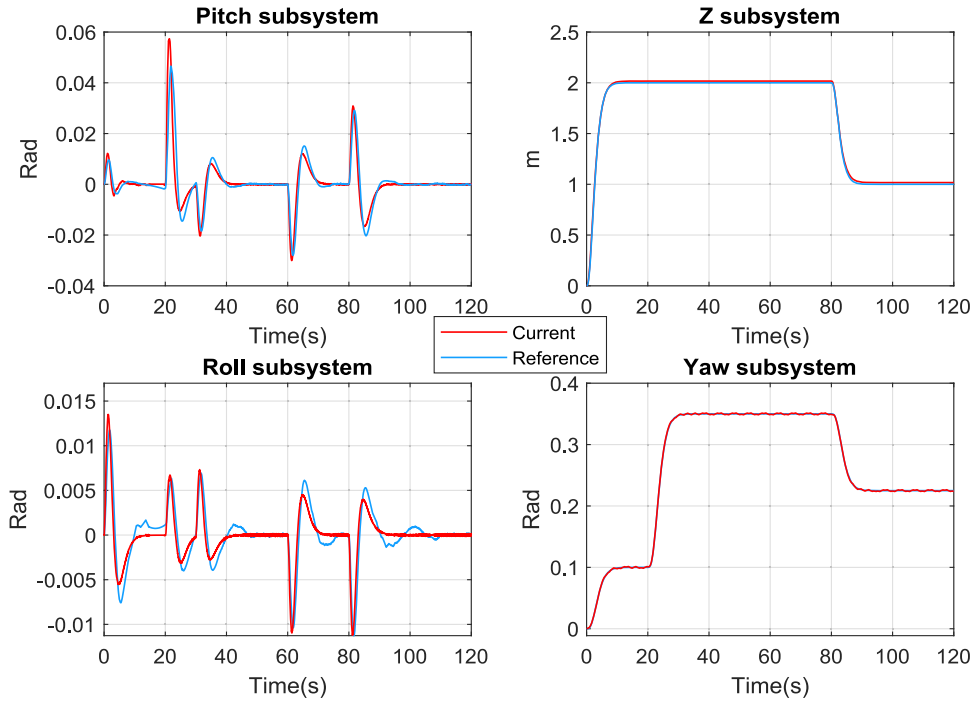


Fig. 13 Fuzzy PID subsystems' responses

membership function in the output. Figure 13 also shows that the fuzzy PID has a longer rise time and some oscillation in the steady state than the SMC-PID in the pitch and roll subsystems. This demonstrates that the fuzzy PID provides a smoother response and improves stability, but it lacks the strong robustness against model uncertainties and nonlinear coupling between the angle and position subsystems that the SMC-PID does.

– **FSMC:**

Figure 14 depicts the FSMC's reaction in all subsystems; in Fig. 16, the FSMC produces smooth control signals with less chattering than the SMC-PID due to the continuous switching signal supplied by the fuzzy logic controller. Furthermore, the FSMC exhibits lower error in steady state than the SMC-PID and fuzzy PID, proving the efficacy of hybridization between fuzzy logic and sliding-mode controller. Furthermore, the FSMC exhibits fewer oscillations and a shorter response time, as well as significant resistance to model uncertainties and nonlinear coupling caused by sliding-mode features.

b. Path tracking performance

Figure 17a, b shows two different path-tracking scenarios applied to the coaxial octorotor using the proposed controllers.

– **SMC-PID:**

The SMC-PID controller responds quickly to the trajectory, with reasonable tracking for the circular trajectory; however, it exhibits some oscillations around the reference Fig. 17a. For the rectangular, the controller follows the trajectory well but exhibits oscillation in the corner, resulting in a rapid direction shift, which is an aggressive move for a drone Fig. 17b. The SMC-PID's weaknesses in the path-tracking task have been identified as a result of the discontinuous switching signal.

– **Fuzzy PID:**

The controller in Fig. 17a performed well in circular trajectory tracking, with a larger offset from the reference than SMC-PID and a slower response to trajectory changes. For the rectangular trajectory in Fig. 17b, it provides better convergence to the reference than the SMC-PID, but in the aggressive maneuvers exhibited in the corners, the controller displays corners that are more round. These results show that the fuzzy logic controller is smooth and has strong convergence, but it lacks the SMC's resilience.

– **FSMC:**

Figure 17a, b show that the FSMC overlaps both paths with quick and smooth convergence, as well as fewer oscillations than the SMC-PID. Figure 17b further illustrates that the FSMC has more sharp corners in the rectangular trajectory with small variations. This highlights the effect of fuzzy logic, which adaptively decreases the discontinuity effect while maintaining the high robustness of the sliding mode, allowing for precise tracking even under nonlinear coupling and model errors.

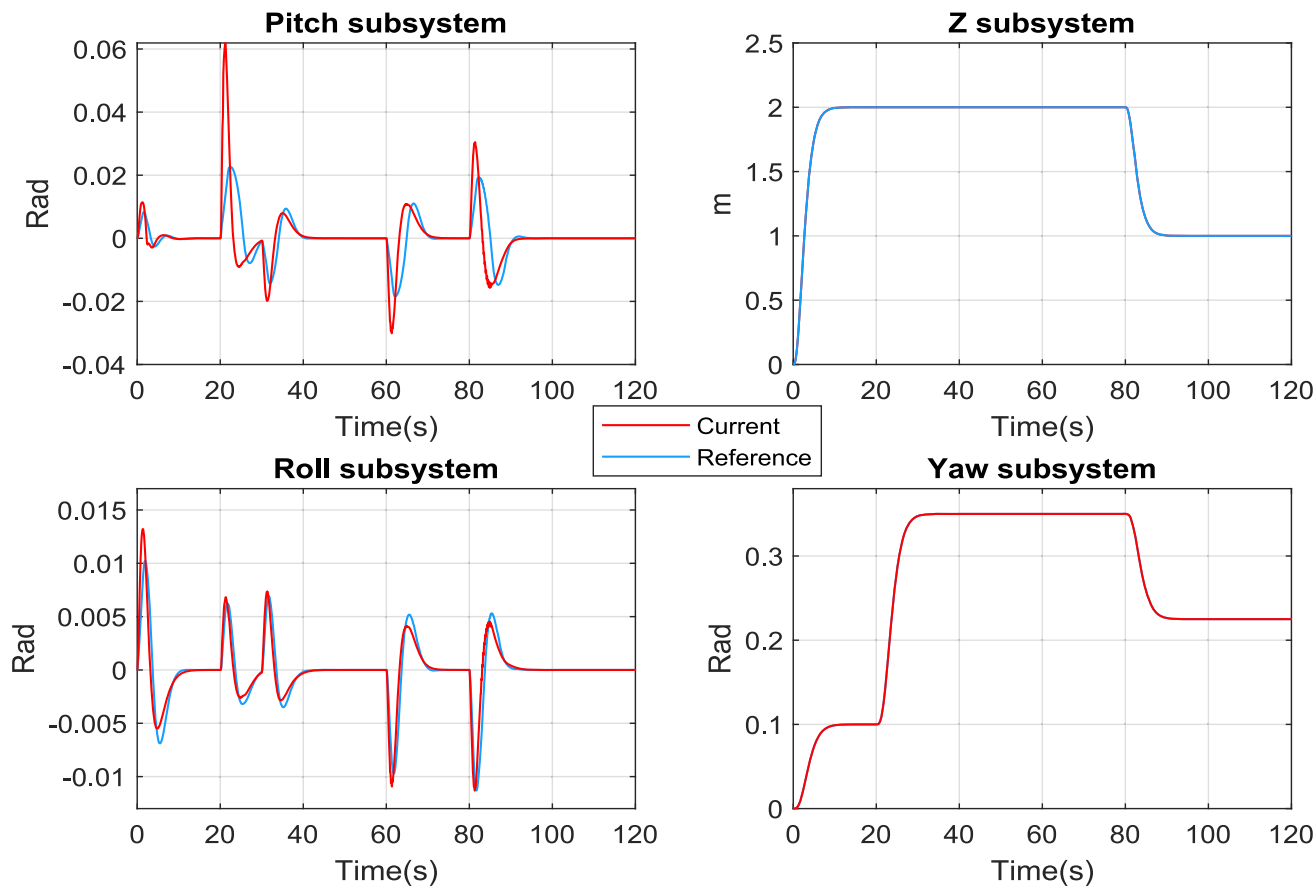


Fig. 14 FSMC subsystems' responses

4.3 Robustness Test Results Analysis

a. The external disturbances

An emulation of real wind disturbances was performed on the developed controllers to assess their resilience. During the route-tracking simulation, an external disturbance of 0.2 rad in pitch angle is applied and sustained for 5 s. The magnitude is roughly equal to 11.50, which is usually regarded as a critical threshold in the literature. Bianchi et al. [33] have shown that a variation of 0.15 to 0.25 rad can affect trajectory tracking under coupled dynamics, particularly in multirotor UAV platforms. Furthermore, in their work on active disturbance rejection control (ADRC) for UAV multirotors [34], the authors justify the use of prolonged pitch deviation as a rigorous test for inner-loop resilience. This insight is completed by incorporating the results reported in [35], where a 0.2 rad deviation in the pitch angle was used as a benchmark to evaluate the LQR controller design.

b. The robustness test and analysis

Figure 18 demonstrates the reaction of the three controllers under an external disturbance. The error performance indices IAE (integral absolute error) and ITAE (integral time absolute error) for the x subsystem across all controllers used

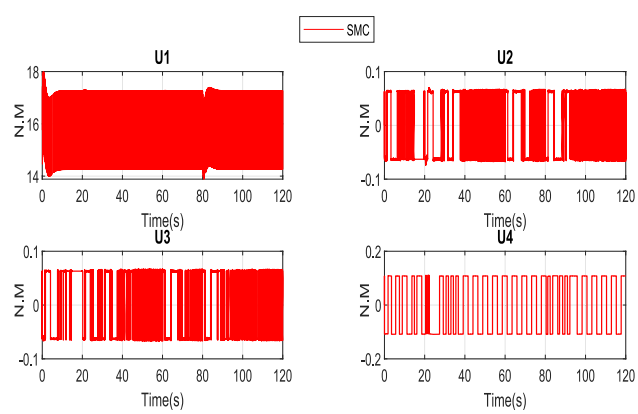


Fig. 15 SMC control signals with chattering

to compare their performance are displayed in Fig. 19. The IAE sums the absolute error over time throughout the path-tracking task under external disturbance. The lower the IAE value, the less the departure from the reference, while the ITAE weights the error by absolute error by time, punishing the stable errors and settling time. The lower the ITAE value, the faster the settling time and the smaller the steady errors.

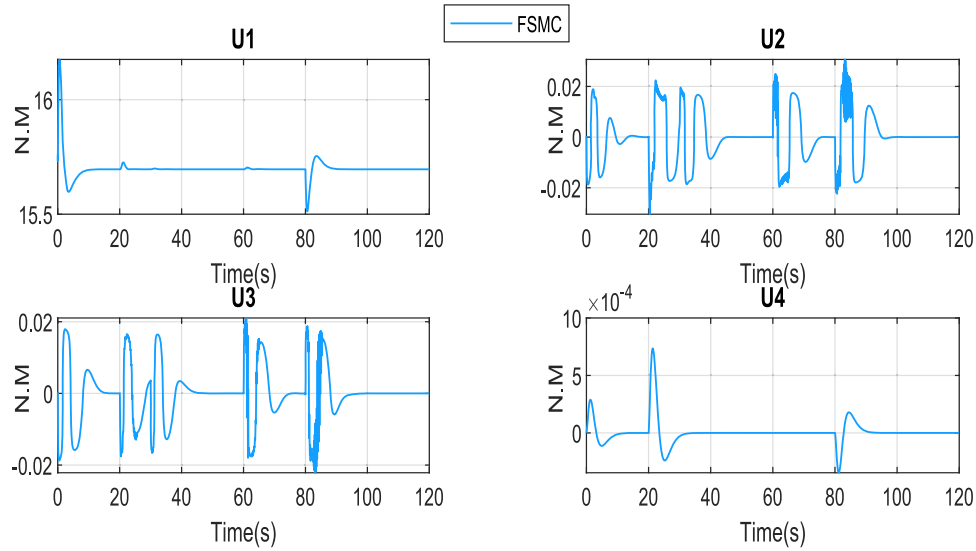


Fig. 16 FSMC control signals chattering elimination

– **SMC-PID:**

The SMC-PID provides a great correction in Fig. 18 due to its robustness, but with significant accumulated error, IAE, and ITAE in Fig. 19a, b, the FSMC indicates that the switching correction may create oscillations and overshoots around the reference point.

– **Fuzzy PID:**

Figure 18 shows that the fuzzy PID responds slowly to external disturbances. Figure 19a, b show that the fuzzy PID has the highest accumulated errors, IAE and ITAE, indicating weaker disturbance rejection and slower repair.

– **FSMC:**

FSMC shows the smallest instantaneous deviation with the fastest return to the reference when a 0.2 rad disturbance is applied in Fig. 18. In addition, the FSMC achieved the lowest accumulated error values, IAE in Fig. 19a and ITAE in Fig. 19a, demonstrating the advantage of the fuzzy logic layer, which makes the switching signal that much smoother, causing less oscillation and overshoot around the reference.

4.4 Quantitative Comparison

From the bar chart showed in Fig. 19a, b, we can compare the performance of the proposed controllers under external disturbance as follows:

- **FSMC**: ITAE = 2.7272× 10^5, IAE = 51.09. (best)
- **SMC-PID**: ITAE = 3.8136× 10^5, IAE = 73.467.
- **FPDC**: ITAE = 5.1022× 10^5, IAE = 117.09. (worst)

The improvements of FSMC versus the SMC-PID and fuzzy PID:

– FSMC reduces **ITAE** by ≈ **28.5%** compared to SMC-PID and by ≈ **46.5%** compared to fuzzy PID.
– FSMC reduces **IAE** by ≈ **30.5%** compared to SMC-PID and by ≈ **56.4%** compared to fuzzy PID.

5 Comparison and Evaluation

Different error performance indices and control effort are utilized to assess the performance of the suggested controllers and select the optimal controller performance in various tracking circumstances. The acquired results demonstrate the great error-tracking performance of the proposed controllers; additionally, we can note that the superior performance indices were attained by using the FSMC. The mathematical representation of the relevant performance indices

– **ITAE** (integral time absolute error):

$$\mathrm{ITAE} = \sum_{k=0}^{N} kT_{\mathrm{s}} \cdot |e(k \cdot T_{\mathrm{s}})|. \tag{41}$$

– **IAE** (integral absolute error):

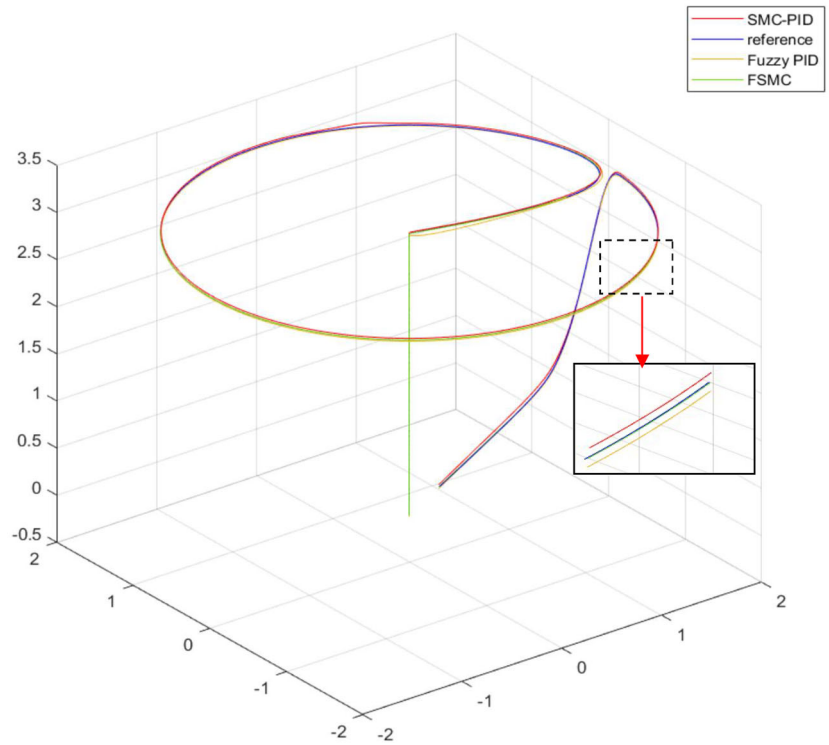
$$\mathrm{IAE} = \sum_{k=0}^{N} |e(k \cdot T_{\mathrm{s}})|\,, \tag{42}$$

where k is discrete time step and T_{s} is the sampling time.

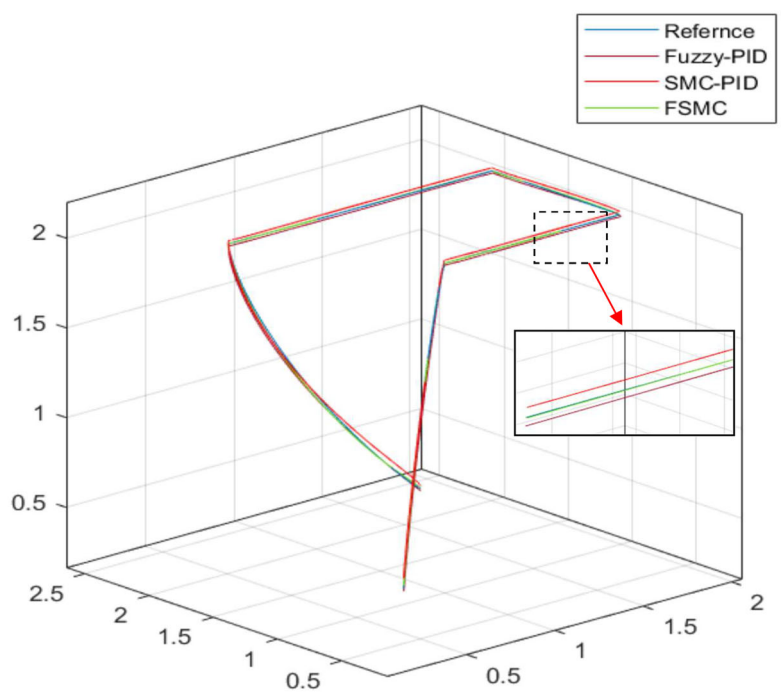
– **CE** (The control effort):

$$\mathrm{CE} = \sum_{i=1}^{N} U_{\mathrm{i}}^{2}, \tag{43}$$

Fig. 17 Obtained results of the tracking tasks



(a) Tracking a circular path



(b) Tracking a rectangular path

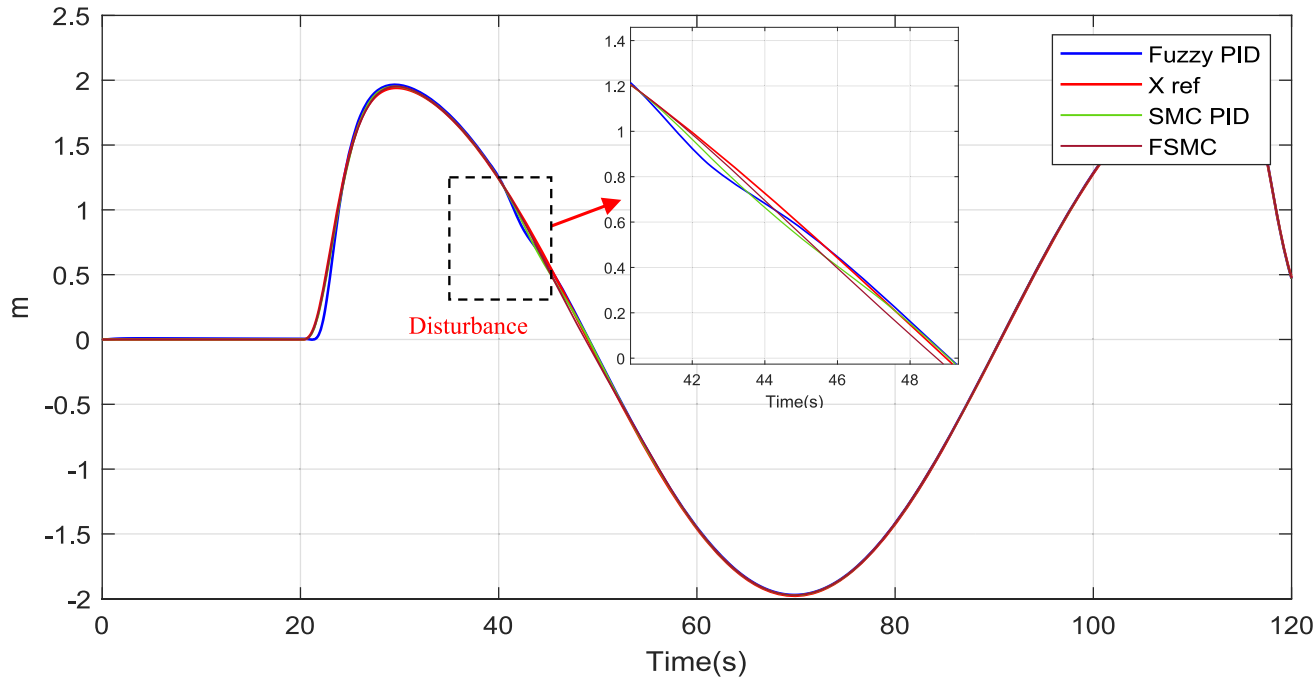


Fig. 18 The controller's responses in the presence of external disturbance on the pitch angle

where $i = [1, 2, 3, 4]$ is the control input number, and U is the control input.

We have calculated these performance indices and the control effort for all controllers on both trajectories (circular and rectangular). Figures 20, 21, 22, 23, 24, and 25 present the error performance indices (ITAE and IAE) and the control effort for both rectangular and circular trajectories, comparing the performance of the three controllers: SMC-PID, fuzzy PID (FPDC), and fuzzy sliding-mode controller (FSMC). These figures provide a comprehensive evaluation of the controllers in terms of tracking accuracy, energy efficiency, and long-term stability.

5.1 Error Performance Indices for the Rectangular Trajectory

a. FSMC Controller:

Figure 20 shows the ITAE error performance index for all controllers; the FSMC achieves the lowest values across all subsystems X, Y, altitude, and yaw. The Integral Time Absolute Error (ITAE) measures the cumulative error weighted by time, demonstrating the controller's ability to minimize steady error from the reference trajectory. The results confirm that the FSMC has the superior long-term tracking performance and fast settling behavior.

In addition, Fig. 21, which shows the IAE error performance index, illustrates that FSMC achieved the smallest cumulative absolute error. This demonstrates its excellent error correction capability and precise transient control. The fuzzy layer within the sliding mode structure enables adaptive gain modulation, allowing the controller to quickly suppress tracking deviations without inducing excessive oscillations.

b. SMC-PID Controller:

With ITAE and IAE values that rank lower than the FSMC, the SMC-PID controller performs admirably. Because of the sliding-mode component, its performance shows robustness, error rejection, and quick transient correction. In contrast to FSMC, the chattering led to increased integrated errors because it reduced the long-term smoothness, especially in the altitude and yaw subsystems. Despite this, SMC-PID fared better than FPDC because of its resilience to model errors and nonlinear coupling.

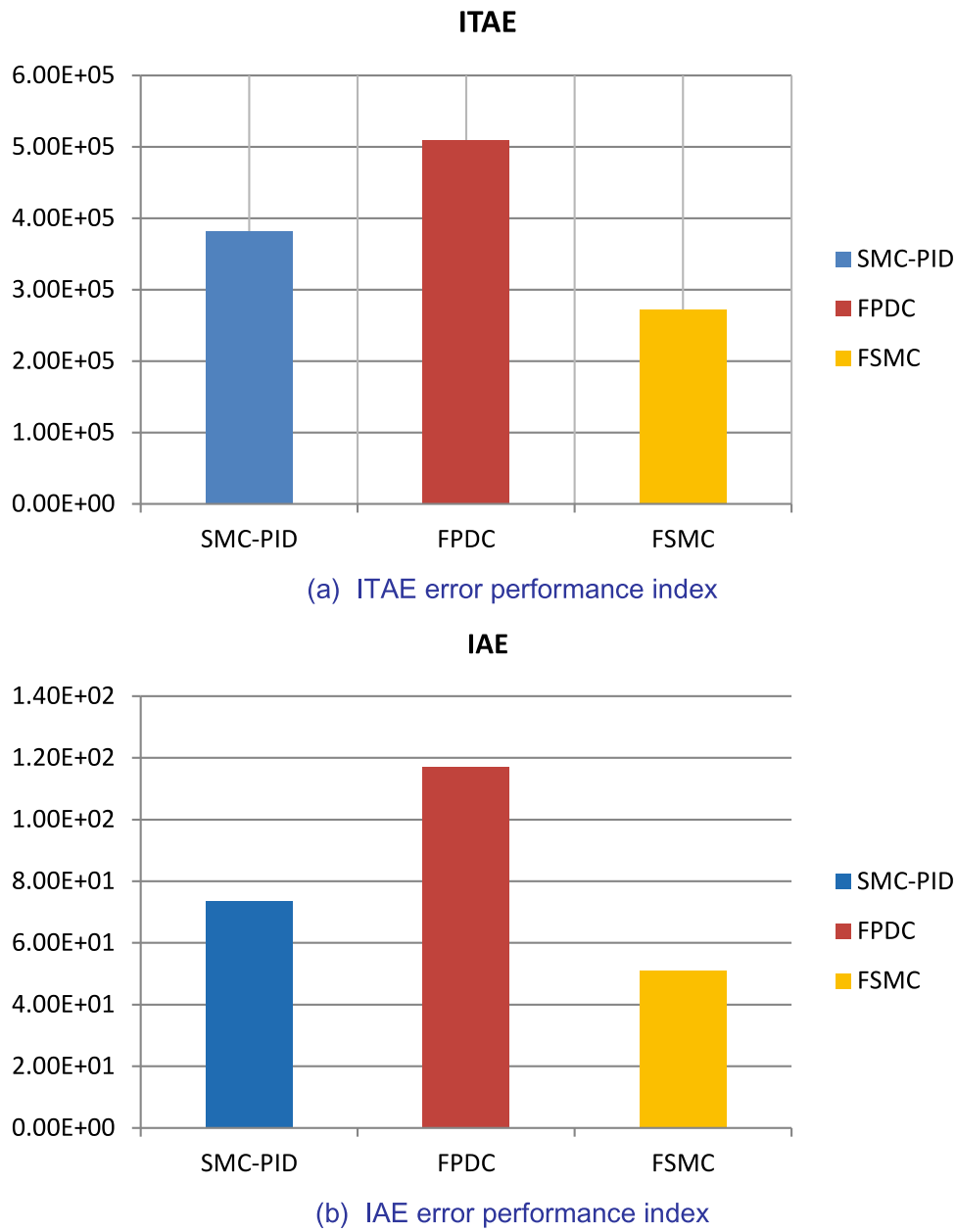
c. FPDC Controller:

Among all subsystems, the fuzzy PID (FPDC) controller achieves the highest ITAE and IAE values, demonstrating the delayed error convergence and poor tracking accuracy. Due to their inability to withstand external disturbances or nonlinear changes, the X and Y subsystems in particular suffer from this.

5.2 Control Effort for the Rectangular Trajectory

Figure 22 shows the control effort for all controllers, which represents the total energy expended by the controllers to maintain the coaxial octorotor on the reference trajectory on the path-tracking task.

Fig. 19 ITAE (**a**) and IAE (**b**) performance indices



- The FSMC achieves the lowest control effort, proving its ability to minimize steady errors with smooth actuator activity. This indicates the energy efficiency and the reduced actuator saturation.
- The SMC-PID shows moderate control effort over all trajectory tracking; its results are lower than fuzzy PID but higher than FSMC. The corrective actions compensate for disturbances, but they cause chattering, which increases the actuators' activity and decreases energy efficiency.
- The fuzzy PID achieves the highest control effort, showing an inefficient energy distribution due to the frequent large corrective actions. This inefficiency related to its higher error indices.

5.3 Error Performance Indices for the Circular Trajectory

For the circular trajectory, the results follow a similar pattern to the rectangular case, emphasizing the consistent superiority of FSMC.

a. FSMC Controller:

Figure 23 (ITAE) and Fig. 24 (IAE) clearly demonstrate that FSMC maintains the lowest integrated errors across all

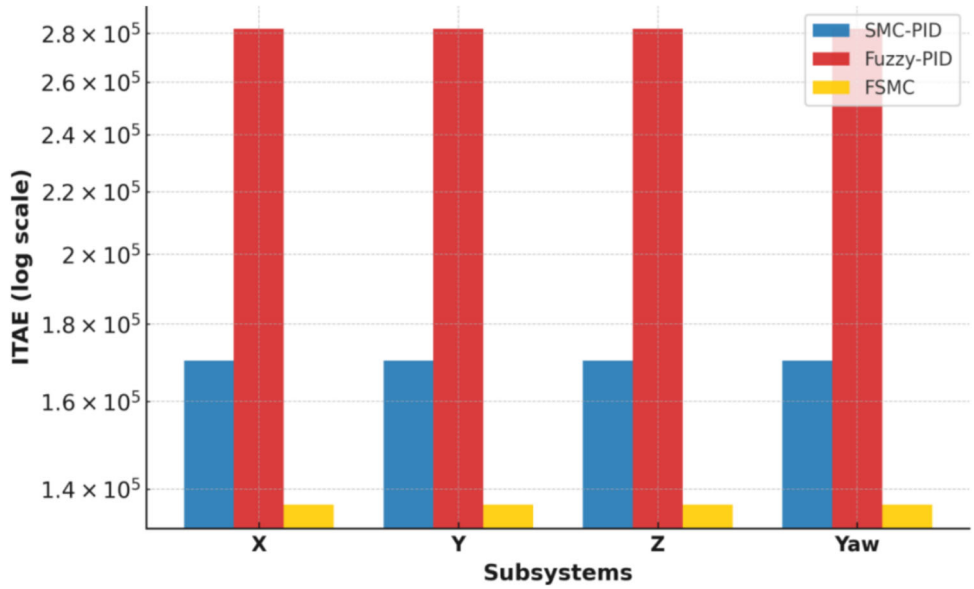


Fig. 20 ITAE for all tracking errors using a rectangular tracking reference

subsystems. The smooth yet adaptive correction mechanism of the fuzzy sliding structure enables the UAV to closely follow the circular reference path with minimal delay or overshoot. Its long-term error minimization and stable convergence confirm robust adaptability to continuous curvature trajectories.

b. SMC-PID Controller:

The SMC-PID controller performs well, achieving smaller ITAE and IAE values than fuzzy PID but higher than FSMC values. The chattering effect caused small oscillations around the reference, especially in the altitude and yaw subsystems. Nevertheless, SMC-PID preserves the good robustness and acceptable accuracy.

c. FPDC Controller:

The fuzzy PID shows the highest ITAE and IAE values during the circular trajectory tracking. The fuzzy PID suffers to maintain the coaxial octorotor tracking precisely the path. especially during continuous directional changes, which reinforce that the fuzzy PID lacks the dynamic uncertainties and nonlinear coupling.

5.4 Control Effort for the Circular Trajectory

Figure 25 depicts the control effort results for all controllers in the circular trajectory-tracking challenge.

- The FSMC delivers the lowest control effort during circular trajectory tracking, demonstrating its capacity to accomplish exact tracking while conserving energy.
- The SMC-PID requires moderate control effort to direct the coaxial octorotor along the circular path. The SMC-PID balanced robustness and performance while spending more energy than FSMC due to the chattering effect, which caused oscillations around the reference.
- The fuzzy PID consumes the most energy during circular trajectory tracking. The larger ITAE and IAE values make the fuzzy PID inefficient in energy consumption and weaker under dynamic uncertainty and nonlinear coupling as compared to the FSMC and SMC-PID.

These findings clearly demonstrate that the hybrid FSMC design provides the optimal compromise between sliding-mode resilience under dynamic uncertainty and nonlinear coupling with the smoothness of fuzzy logic. This hybridization provides the FSMC with precision and control smoothness, making it the best controller for autonomous flight of coaxial octorotor in both structured and disturbed situations in the scenes used in this research.

6 Conclusion

This study presents and compares three fuzzy-based nonlinear control strategies, namely fuzzy PD controller (FPDC), sliding-mode PID (SMC-PID), and fuzzy sliding-mode controller (FSMC), for altitude and attitude control of a six-degree-of-freedom (6-DOF) coaxial octorotor UAV. The main objective was to evaluate their performance in trajectory tracking, disturbance rejection, and energy efficiency using a complete nonlinear dynamic model of the system.

The proposed FSMC integrates a type-1 Mamdani fuzzy logic controller into the switching part of the sliding-mode control. The fuzzy inference mechanism generates a continuous control signal that replaces the discontinuous

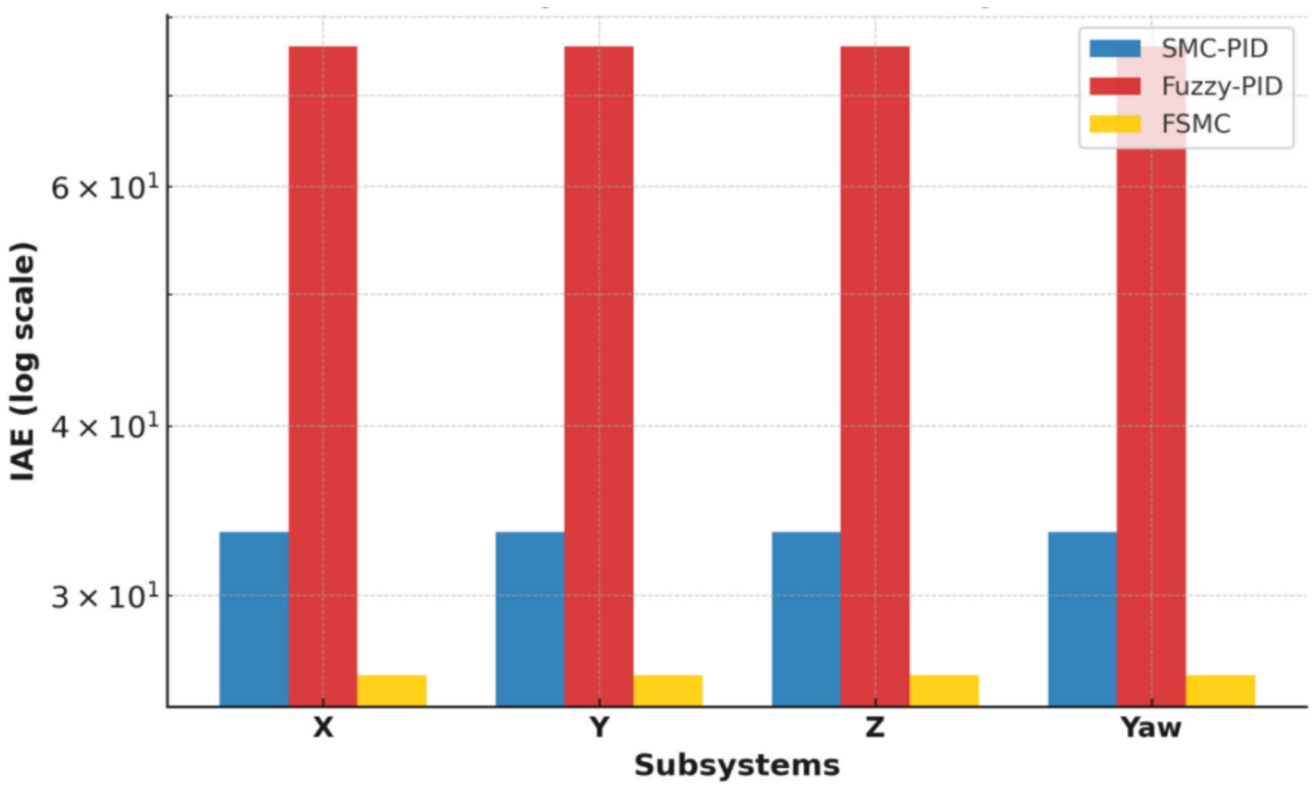


Fig. 21 IAE for all tracking errors using a rectangular tracking reference

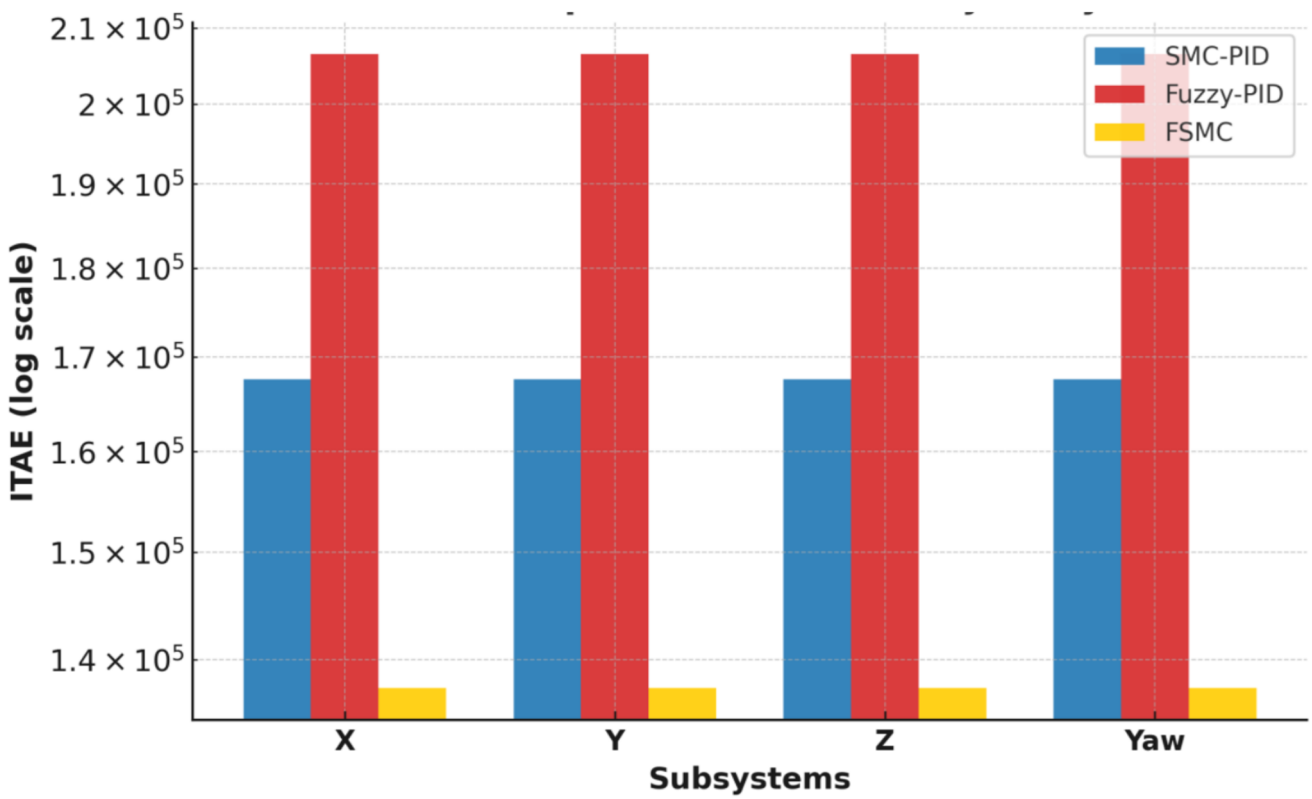


Fig. 22 ITAE for all tracking errors using a circular tracking reference

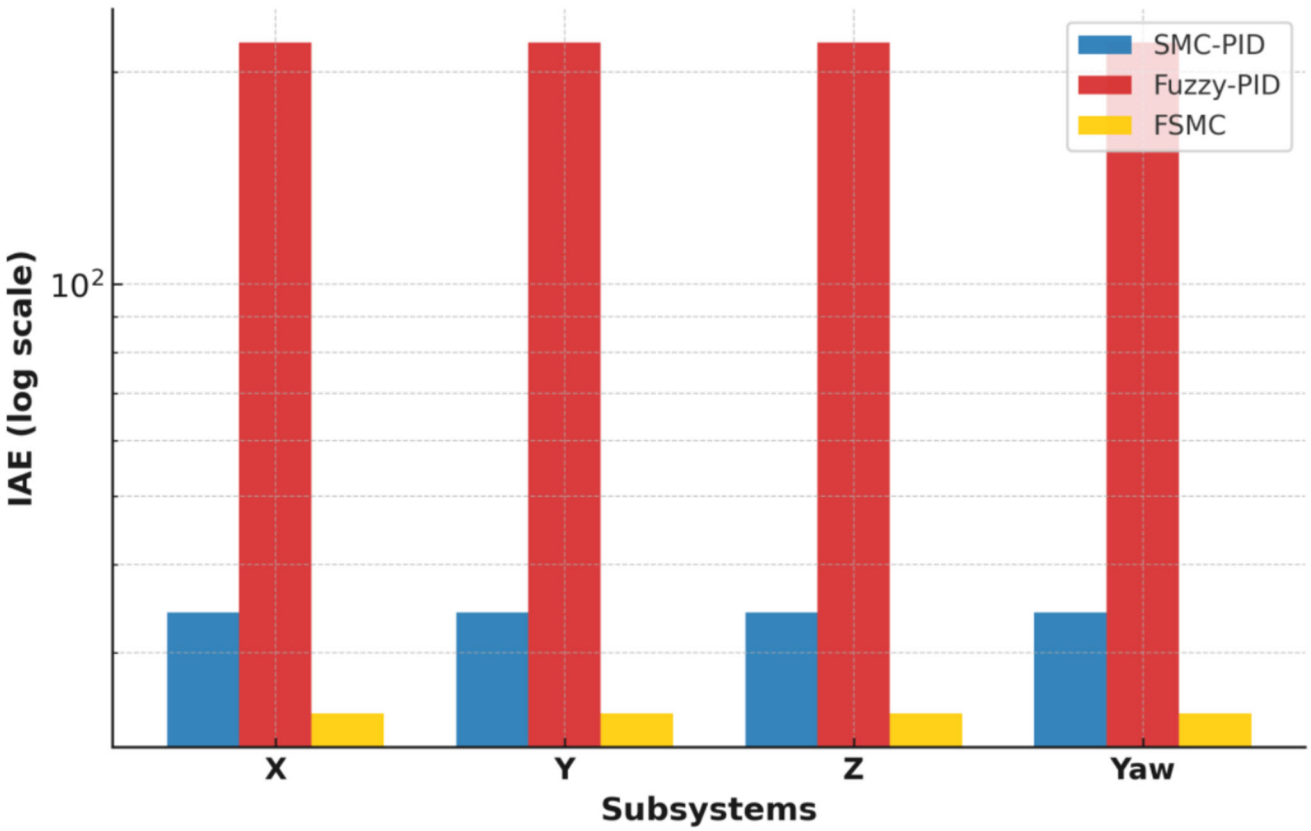
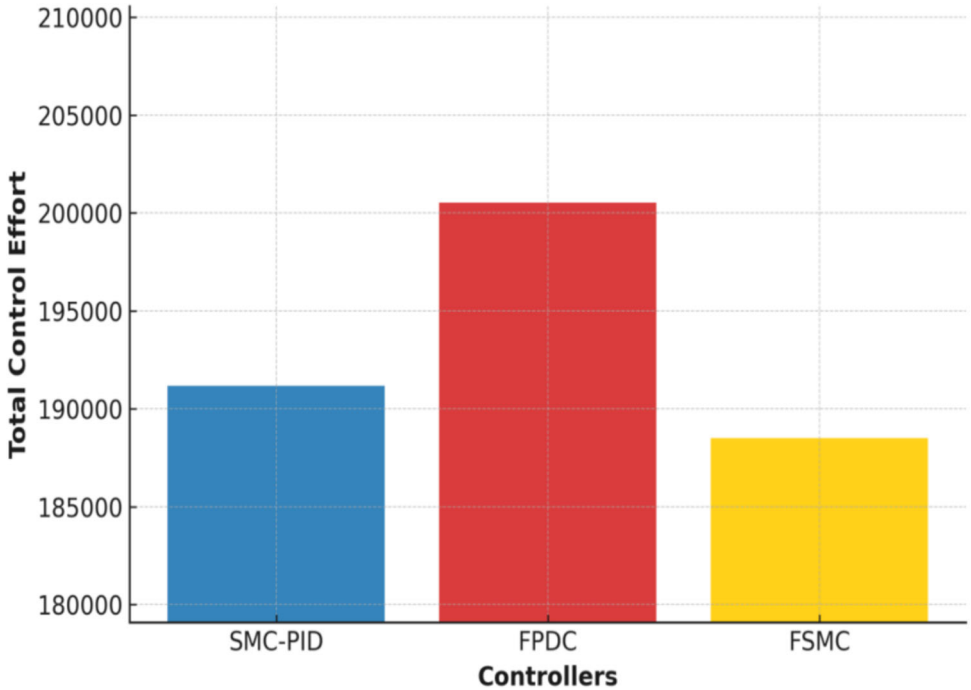


Fig. 23 IAE for all tracking errors using a circular tracking reference

Fig. 24 The control effort using a rectangular tracking reference



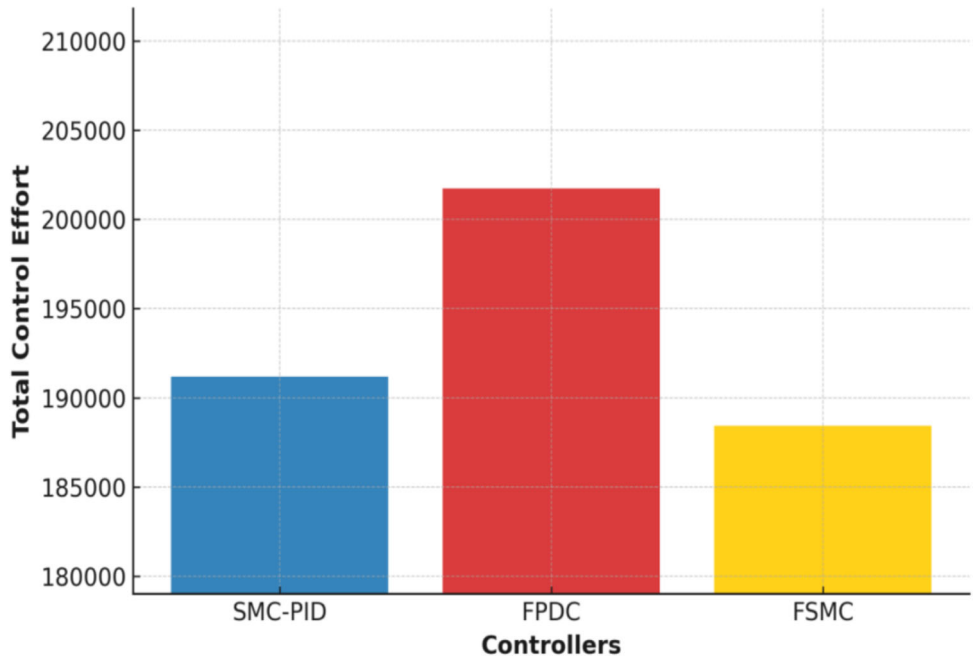


Fig. 25 The control effort using a circular tracking reference

sign function, which effectively suppresses chattering while maintaining robustness and stability. The FSMC's fuzzy rules and membership functions were tuned through simulation to ensure smooth transient behavior and fast error convergence. Simulation tests were conducted for both rectangular and circular trajectories under nominal and disturbed conditions. The obtained results show that FSMC consistently provides the smallest Integral of Absolute Error (IAE) and Integral of Time-weighted Absolute Error (ITAE) indices, confirming superior tracking accuracy and faster stabilization. It also demonstrates the lowest control effort compared with FPDC and SMC-PID, which translates to reduced energy consumption. In the presence of external wind disturbances, FSMC maintains trajectory tracking with minimal deviation, proving its robustness and adaptability. In contrast, FPDC provides acceptable performance with simpler structure and lower computational demand, making it suitable for low-cost systems where extreme precision is not required. SMC-PID maintains strong robustness to disturbances but generates larger oscillations and higher energy usage due to the discontinuous switching term. These comparative findings show that FSMC achieves the most balanced performance in terms of accuracy, robustness, and energy efficiency.

Overall, the FSMC structure offers a promising solution for the control of coaxial octorotor UAVs, combining the advantages of fuzzy inference and sliding-mode control. Its ability to ensure smooth response, reduced chattering, and reliable disturbance rejection confirms its suitability for real-time implementation in autonomous flight applications.

Future work will focus on adaptive tuning of fuzzy and sliding parameters using optimization and machine learning algorithms. Additional research will also include experimental validation on a real coaxial octorotor platform and extension of the control design to handle actuator faults and sensor noise for practical field deployment.

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Declarations

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Ethical Approval Not applicable.

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Appendix

Algorithm 1: Sliding Mode with PID Control

Inputs:
- Desired states: (x_d, y_d, z_d, ϕ_d, θ_d, ψ_d)
- Current states : (x, y, z, ϕ, θ, ψ)
- Sliding-mode control parameters: (λ,K,C)
- PID gains: (K_p,K_d,K_i)

Initialization:
- Assign initial errors: e_x, e_y, e_z, e_ϕ, e_θ, e_ψ.
- Define sliding surfaces: s_x, s_y, s_z, s_ϕ, s_θ, s_ψ.

Processing:

1. Compute the error for each state:
$$ex = xd - x, \quad ey = yd - y, \quad ez = zd - ze$$
$$e\phi = \phi d - \phi, \quad e\theta = \theta d - \theta, \quad e\psi = \psi d - \psi e$$
2. Define the sliding surface for each subsystem:
$$S_Z = e^{\cdot}{}_Z + \lambda_Z e_Z S_\phi = e^{\cdot}{}_\phi + \lambda_\phi e_\phi$$
$$S_\theta = e^{\cdot}{}_\theta + \lambda_\theta e_\theta S_\psi = e^{\cdot}{}_\psi + \lambda_\psi e_\psi$$
3. For each subsystem:

- Altitude control (U1) **:**
- Define dynamics:
$$\ddot{z} = \frac{\cos\varphi\cos\theta}{m} u_1 - \frac{K_3}{m}\dot{z} - g$$
- Compute control input:
$$U_1 = \frac{1}{\cos\varphi.\cos\theta}\left[K_3 + m_g + \ddot{z}_d.m + \dot{e}.c.m + sgn\,(s)\,.K\right]$$

- Pitch control (U2) **:**
- Define dynamics:
$$\ddot{\varphi} = \frac{(I_y - I_z)}{I_x}\dot{\theta}\dot{\psi} - \frac{K_4}{I_x}\dot{\varphi}^2 - \frac{J_r\bar{\Omega}}{I_y}\dot{\varphi} + \frac{1}{I_x}u_2.$$
- Compute control input :
$$U_2 = I_x\left[\dot{e}_\varphi.c + \ddot{\varphi}_d - \left(\dot{\theta}.\dot{\psi}.A_1 - \dot{\varphi}^2.A_2 - A_3.\dot{\varphi}\right) + sgn\,(s)\,.K\right]$$
$$A_2 = \frac{K_4}{I_x}A_1 = \frac{(I_y - I_z)}{I_x}A_3 = \frac{J_r\bar{\Omega}}{I_y}$$

- Roll control (U3)**:**
- Define dynamics:
$$\ddot{\theta} = \frac{(I_z - I_x)}{I_y}\dot{\varphi}\dot{\psi} - \frac{K_5}{I_y}\dot{\theta}^2 - \frac{J_r\bar{\Omega}}{I_x}\dot{\theta} + \frac{1}{I_y}u_3$$
- Compute control input:
$$U_3 = I_y\left[\dot{e}_\theta.c + \ddot{\theta}_d - \left(\dot{\varphi}.\dot{\psi}.A_5 - \dot{\theta}^2.A_6 - A_7.\dot{\theta}\right) + sgn\,(s)\,.K\right]$$
$$A_6 = \frac{K_5}{I_y}A_5 = \frac{(I_z - I_x)}{I_y}A_7 = \frac{J_r\bar{\Omega}}{I_x}$$

- Yaw control (U4)**:**
- Define dynamics:
$$\ddot{\psi} = \frac{(I_x - I_y)}{I_z}\dot{\theta}\dot{\varphi} - \frac{K_6}{I_z}\dot{\psi}^2 + \frac{1}{I_z}u_4$$
- Compute control input:
$$U_4 = I_z\left[\dot{e}_\psi.c + \ddot{\psi}_d - \left(\dot{\varphi}.\dot{\theta}.A_9 - \dot{\psi}^2.A_8\right) + sgn\,(s)\,.K\right]$$
$$A_8 = \frac{K_6}{I_z}A_9 = \frac{(I_x - I_y)}{I_z}$$

4. Use PID controllers for X and Y subsystems to generate θ_d and ϕ_d:
- Compute desired pitch :
$$\theta_d = K_p.e_y + K_i.\int e_y + K_d.\frac{de_y}{dt}$$
- Compute desired roll :
$$\varphi_d = K_p.e_x + K_i.\int e_x + K_d.\frac{de_x}{dt}$$
5. Repeat until all states converge to their desired values.

Outputs :
. Control inputs : (U1, U2, U3, U4).

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